

Submersible Motor Pump

Amarex KRT

60 Hz

Type Series Booklet



Legal information/Copyright

Type Series Booklet Amarex KRT

All rights reserved. The contents provided herein must neither be distributed, copied, reproduced, edited or processed for any other purpose, nor otherwise transmitted, published or made available to a third party without the manufacturer's express written consent.

Subject to technical modification without prior notice.

© KSB Aktiengesellschaft, Frankenthal 28.10.2016

Contents

Waste Water	4
Submersible Motor Pump	4
Amarex KRT	4
Main applications	4
Fluids handled	4
Operating data	4
Designation	4
Design details	5
Materials	6
Product benefits	7
FluidFuture energy-efficiency concept by KSB	7
Acceptance tests and warranty	7
Selection information	7
Overview of product features / selection tables	8
Overview of product features	8
Impellers	13
Table of fluids handled	14
Shaft seal	16
Technical data	17
Mass moments of inertia per motor	23
Selection charts	25
Amarex KRT S-max, n = 3500 rpm	25
Amarex KRT S, n = 3500 rpm	26
Amarex KRT F-max, n = 3500/1750 rpm	27
Amarex KRT F, n = 3500/1750/1160 rpm	28
Amarex KRT E-max, n = 3500/1750 rpm	29
Amarex KRT E, n = 1750/1160 rpm	30
Amarex KRT D, n = 1750/1160/875 rpm	31
Amarex KRT K-max, n = 1750/1160 rpm	32
Amarex KRT K 40-.../80-.../100-.../150-.../200-.../250-..., n = 3500/1750 rpm	33
Amarex KRT K 100-.../150-.../151-.../200-.../250-.../300-.../350-..., n = 1160 rpm	34
Amarex KRT K 200-.../250-.../300-.../350-.../400-.../500-..., n = 875 rpm	35
Amarex KRT K 350-.../500-.../600-.../700-..., n = 700/585/500 rpm	36
Installation types	37
Scope of supply	37
General assembly drawings with list of components	38
Amarex KRT, motor type 1	38
Amarex KRT, motor type 2	39
Amarex KRT, motor type 3	40
Amarex KRT, motor type 4, installation types S and P	41
Amarex KRT, motor type 4, installation types K and D	42

Waste Water

Submersible Motor Pump

Amarex KRT



Main applications

- Waste water management
- Service water supply systems
- Disposal
- Sewage treatment plants
- Sludge disposal

Fluids handled

- Sewage containing feces
- Activated sludge
- Digested sludge
- Raw sludge
- Fluids containing gas
- Industrial waste water

Operating data

Operating properties

Characteristic		Value
Flow rate	Q [US.gpm]	≤ 45000
	Q [l/s]	≤ 2800
Head	H [ft]	≤ 330
	H [m]	≤ 100
Fluid temperature	T [°F]	≤ 140
	T [°C]	≤ 60

Characteristic		Value
Motor rating	P ₂ [hp]	≤ 1140
	P ₂ [kW]	≤ 850

Designation

Example: Amarex KRT K 150-401/80 4 UN G-S IE3

Designation key

Code	Description
Amarex KRT	Type series
K	Impeller type (⇒ Page 5)
D	Open, diagonal single-channel impeller
E/E-max	Closed single-channel impeller
F/F-max	Free-flow impeller
K/K-max	Closed multi-channel impeller
S	Impeller with cutter
150	Nominal discharge nozzle diameter [mm]
401	Maximum nominal impeller diameter [mm]
80	Motor size
4	Number of poles
	2, 4, 6, 8, 10, 12
UN	Motor version
U/UN/UE	Not explosion-proof, for fluid temperatures of up to 104 °F [40 °C]
W/WN/WE	Not explosion-proof, for fluid temperatures of up to 140 °F [60 °C]
X/XN/XE	Explosion-proof, for fluid temperatures of up to 104 °F [40 °C]
Z/ZN/ZE	Explosion-proof, for fluid temperatures of up to 140 °F [60 °C]
G	Material variant
G	Standard design, gray cast iron
G1	Like G, impeller made of duplex stainless steel
G2	Like G, impeller made of white cast iron
GH	Like G, impeller and discharge cover made of white cast iron
H	Wetted components made of white cast iron
C1	Wetted components made of duplex stainless steel, mechanical seal with elastomer bellows, screws/bolts made of A4/A 276 Type 316 Ti
C2	Wetted components made of duplex stainless steel, mechanical seal with covered spring, screws/bolts made of 1.4462/A 182 F51
S	Type of installation (⇒ Page 37)
S	Stationary wet installation with guide cable or guide rail installation (without cooling jacket)
D	Stationary dry installation, vertical
P	Transportable model for wet installation
K	Stationary wet installation with guide cable or guide rail installation

Code	Description	
	H	Stationary dry installation, horizontal
IE3	Motor efficiency classification	
	1)	No efficiency classification
	IE2, IE3	Efficiency classification ²⁾

Design details

Design

- Fully floodable submersible motor pump
- Not self-priming
- Close-coupled design

Drive

- Three-phase asynchronous squirrel-cage motor
- Version with explosion protection to NEC 500: Explosion-proof for Class I, Division 1, Groups C and D, T3, hazardous (classified) locations.

Shaft seal

- Two bi-directional mechanical seals in tandem arrangement, with liquid reservoir

- Pumps with reinforced bearings: with leakage chamber

Impeller type

- Various application-oriented impeller types

Bearings

Standard bearings:

- Grease-packed bearings sealed for life
- Maintenance-free

Reinforced bearings:

Drive end:

- Grease-packed bearings sealed for life
- Maintenance-free

Pump end:

- Grease-packed bearings
- Can be re-lubricated

¹⁾ Blank

²⁾ IEC 60034-30 standard not binding for submersible motor pumps. Efficiencies calculated/determined according to the measurement method specified in IEC 60034-2. The marking is used for submersible motors that achieve efficiency levels similar to those of standardized motors acc. to the IEC 60034-30 standard.

Materials

Overview of available materials

Part No.	Description	Material variant						
		G	G1	G2	GH	H	C1	C2
Pump set								
101	Pump casing	EN-GJL-250			EN-GJN-HB555		1.4517	
135	Wear plate ³⁾	EN-GJL-250		-				
163	Discharge cover	EN-GJL-250			EN-GJN-HB555		1.4517	
210	Shaft	1.4021/C45+N (⇒ Page 8)					1.4021/1.4462/C45+N (⇒ Page 8)	
230	Impeller ⁴⁾	EN-GJL-250	1.4517	EN-GJN-HB555			1.4517	
330	Bearing bracket	EN-GJL-250					1.4517	
350	Bearing housing	EN-GJL-250					1.4517/EN-GJL-250	
412	O-ring	Nitrile butadiene rubber NBR						Viton (FKM)
433.01	Mechanical seal (drive end)	Carbon/SiC						
433.02	Mechanical seal (pump end)	SiC/SiC						
502	Casing wear ring ⁵⁾	EN-GJL-250			VG 434			
66-2	Cooling jacket	1.4571			-			
811	Motor housing	EN-GJL-250					1.4517	
824	Power cable	(⇒ Page 12)						
900	Screws	A4 ⁶⁾						1.4462
Installation parts								
572	Guide cable suspension bracket	1.4571 to DN 200; EN-GJL-250 from size 200-500						1.4571
59-24	Guide cable	1.4401						1.4401/ Tefzel
72-1	Flanged bend	EN-GJL-250			EN-GJN-HB555		1.4517	
732	Claw	EN-GJL-250 or EN-GJS-400-15/EN-GJS-500-7					1.4517	
885	Lifting chain/rope	Lifting chain: 1.4404 Lifting rope: polyamide/polypropylene					Lifting rope: polypropylene	
892	Foot plate / feet	1.0038 + Z					1.4571	1.4517/ 1.4462
894	Bracket	1.4571 to DN 200; 1.0038 + Z from size 200-500						1.4571

Comparison of materials

EN	ASTM
EN-GJL-250	A 48 Class 35 B
EN-GJS-400-15	A 536 Class 60-40-18
EN-GJN-HB555	A 532 Class II Type B (15 % Cr-Mo)
1.4517	A 890 CD4MCuN
1.4021	A 276 Type 410
1.4401	A 276 Type 316 L
1.4462	A 182 F51
VG 434	-
1.4571	A 276 Type 316 Ti
C45+N	A 576 Gr. 1045
NBR	NBR
FKM	FKM

Description of materials

Gray cast iron EN-GJL-250 (lamellar graphite cast iron):

Lamellar graphite cast iron to EN 1561 is the most widely used cast material for handling municipal sewage, waste water and sludges as well as stormwater and surface water. It is suitable for neutral fluids which are only slightly aggressive and cause little wear. The pH value should be ≥ 6.5 , the sand content ≤ 0.5 g/l.

Duplex stainless steel (1.4517 or technically equivalent material)

This type of cast steel is resistant to cavitation, has excellent strength values and is used for high circumferential speeds. An excellent resistance to pitting corrosion makes ferritic-austenitic stainless steel a popular choice for pumping acidic waste with a high chloride content, as well as seawater and brackish water. Thanks to its good chemical resistance, e.g. against waste water containing phosphorous and sulphuric acid, this material is used in a wide range of applications in the chemical industry and process engineering. Pumps made of duplex stainless steel have a very long service life, even when handling brines, chemical waste water (pH 1 - 12), gray water and landfill leachate.

Wear-resistant white cast iron ((EN-GJN-HB555 [XCR14] or technically equivalent material)

This is a wear-resistant white cast iron for highly abrasive fluids containing sand, ash or iron ore sinter. It has a Rockwell hardness of 61.5 to 68, which is higher than that of hardened chrome steel. Owing to its hardness, the chromium-molybdenum alloy cast iron features a notably higher wear resistance than EN-GJL-250 gray cast iron and other cast materials.

- 3) For D impeller
 4) D impeller: EN-GJL-250, with hardened edges
 5) For E impeller and K impeller
 6) Equivalent to 1.4571

Product benefits

- Absolutely water-tight resin-sealed cable entries prevent any water from entering the motor – even in the event of a damaged connection cable.
- Reliable operation ensured by moisture sensors signaling any ingress of moisture into the motor
- Reliable operation ensured by sensors monitoring the motor temperature and preventing overheating
- Non-clogging low-maintenance design with large free passages reduces clogging risk and, consequently, maintenance work.
- Optimum hydraulic and energy efficiency by high-efficiency motors and variable hydraulic systems

Material variants C1 and C2:

- Long service life due to corrosion-resistant wetted components made of stainless steel

FluidFuture energy-efficiency concept by KSB



www.ksb.com/fluidfuture

Acceptance tests and warranty

Functional test

- Every pump undergoes functional testing to KSB standard ZN 56535.
- Operating data is guaranteed to DIN EN ISO 9906/2/2B or Hydraulic Institute Level A/B.

Acceptance tests

- Acceptance testing to ISO/DIN or comparable standards is available against a surcharge.
- Acceptance tests to Hydraulic Institute available on request.

Warranty

- Quality is assured by means of an audited and certified quality assurance system to DIN EN ISO 9001.

Selection information

- The indicated heads and performance data apply to material variant G, for fluids handled with a density $\rho = 1 \text{ kg/dm}^3$ and a kinematic viscosity $\nu \leq 20 \text{ mm}^2/\text{s}$.

- For hydraulic acceptance tests of material variants other than "G" reduce the documented efficiencies by 2 percentage points.

Impeller type

- S, F, E and D impellers can only be supplied with the documented impeller diameters. Indicate the pump set designation and the impeller diameter in the purchase order.
- K impellers are trimmed to the duty point. Indicate the H/Q data or the impeller diameter in the purchase order. In the hydraulic selection program, the impeller diameter is automatically computed based on the H/Q data and added to the designation of the pump set.

Pump input power

- Adjust the power input to the density of the fluid handled:

$$P_2 (\text{required}) = \rho [\text{kg/dm}^3] (\text{fluid handled}) \times P_2 (\text{documented})$$
- Select the operating point with the largest power input within an operating range. Select a motor size providing a power reserve to compensate for the tolerances in the system characteristic / pump characteristic.

Recommended motor power reserve⁷⁾

P ₂		Reserve	
[hp]	[kW]	Mains operation	With frequency inverter
≤ 40	≤ 30	10 %	15 %
> 40	> 30	5 %	10 %

- For installation types K and D (with cooling jacket) an additional power reserve of 2 hp [1.5 kW] must always be added for the cooling circuit.
- The motors are suitable for operation on a frequency inverter.
- The service factor for all motors equals 1.15.

i In the case of waste water, too low a flow velocity in the discharge line will lead to clogging and increased wear. The flow velocity in the vertical riser must not fall below 6.56 ft/s [2 m/s].

i In the case of waste water, too low a circumferential speed of the impeller will lead to clogging of the hydraulic system (operation on frequency inverter). A minimum circumferential speed (measured at the outside diameter of the impeller) of 39.4 ft/s [12 m/s] must be observed.⁸⁾

⁷⁾ If larger power reserves are stipulated by local regulations, these larger reserves must be provided.

⁸⁾ For F impellers, a circumferential speed below 39.4 ft/s [12 m/s] is permissible.

Overview of product features / selection tables

Overview of product features

Material variants G, G1, G2, GH

Feature	Motors						
2-pole	3 2E ... 26 2 E	-	37 2 ... 55 2	-	-	-	-
4-pole	2 4E ... 22 4 E	23 4 ... 29 4	35 4 ... 65 4	35 4 N ... 80 4 N	95 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6-pole	7 6E ... 18 6 E	20 6	32 6 ... 50 6	32 6 N ... 60 6 N	80 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8-pole	-	10 8 ... 17 8	26 8 ... 35 8	26 8 N ... 50 8 N	75 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10-pole	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12-pole	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Suction flange	9)						
Shaft material							
Shaft	A 276 Type 420			A 576 Gr. 1045	A 276 Type 420		
Shaft protecting sleeve	-			A 276 Type 420			
Bearings	Grease-packed rolling element bearings sealed for life		Grease-packed rolling element bearings sealed for life ¹⁰⁾	Pump end: re-greasable rolling element bearings Drive end: grease-packed rolling element bearings sealed for life			
Explosion protection							
Versions U, W	Not explosion-proof						
Versions X, Z	 Or  Explosion-proof Class I, Division I, Groups C & D, T3						-
Motor							
Starting method	DOL, soft starter, frequency inverter or star-delta			DOL, soft starter, frequency inverter			
Voltage	460 V ¹¹⁾						460 V ¹²⁾
Cooling	Cooled by surrounding fluid ¹³⁾	Cooled by surrounding fluid		Cooled by surrounding fluid or via cooling jacket			
Immersion depth	100 ft [30 m] max.						
Power cable							
Type	See table "Overview of power cables"						
Length	30 ft [10 m] ¹⁴⁾						
Cable entry	Absolutely watertight						
Sealing elements							
Elastomer seals	Nitrile butadiene rubber NBR ¹⁵⁾						

9) Drilled to ANSI B16.1 Class 125; not available for all sizes; on request

10) For D impeller: re-greasable rolling element bearings at the pump end, grease-packed rolling element bearings sealed for life at the drive end

11) Optional: 200 V, 230 V, 380 V, 575 V

12) Optional: 380 V, 575 V

13) Optional: air cooling for motors 11 2E ... 26 2E; 7 4E ... 22 4E; 7 6E ... 18 6E;

14) Optional: up to 164 ft [50 m]

Feature	Motors						
2-pole	3 2E ... 26 2 E	-	37 2 ... 55 2	-	-	-	-
4-pole	2 4E ... 22 4 E	23 4 ... 29 4	35 4 ... 65 4	35 4 N ... 80 4 N	95 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6-pole	7 6E ... 18 6 E	20 6	32 6 ... 50 6	32 6 N ... 60 6 N	80 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8-pole	-	10 8 ... 17 8	26 8 ... 35 8	26 8 N ... 50 8 N	75 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10-pole	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12-pole	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
Shaft seal	Bellows-type mechanical seal ¹⁶⁾						Mechanical seal with covered spring
Monitoring equipment							
Winding temperature version U, W; installation type S, P	Temperature switch (bimetal) in the winding						
Winding temperature version X, Z; installation type S, P	Temperature switch (bimetal) in the winding, plus PTC for explosion protection ¹⁷⁾	Temperature switch (bimetal) in the winding, plus PTC for explosion protection					-
Winding temperature; installation type D, H, K	PTC	-		PTC			
Coolant temperature; installation type D, K	-			PTC			
Bearing temperature	-	_18)		Pt100 at the pump end ¹⁹⁾			Pt100 at the pump and drive end
Leakage inside the motor	Leakage sensor in the motor space						
Mechanical seal leakage	-		Float switch in the leakage area for D impeller	Float switch in leakage area			
Vibration sensor	-		-	_20)			
Coating	Environmentally friendly KSB standard coating, color RAL 5002 ²¹⁾						
Maximum fluid and ambient temperature							
Version U	104 °F [40 °C]						
Version X	104 °F [40 °C]						-
Versions W, Z	140 °F [60 °C]						-
Tests/inspections							
Hydraulic system	KSB standard (ZN 56525) ²²⁾						

¹⁵⁾ Optional: Viton = fluorocarbon rubber FPM

¹⁶⁾ Optional: mechanical seal with covered spring

¹⁷⁾ Motors 3 2E 2 4E, 3 4E: temperature switch (bimetal) in the winding, plus temperature switch (bimetal) for explosion protection

¹⁸⁾ Optional: Pt100 at the pump end

¹⁹⁾ Optional: Pt100 at the drive end

²⁰⁾ Optional: internal vibration sensor

²¹⁾ Optional: 250 µm two-component epoxy coating

Feature	Motors						
2-pole	3 2E ... 26 2 E	-	37 2 ... 55 2	-	-	-	-
4-pole	2 4E ... 22 4 E	23 4 ... 29 4	35 4 ... 65 4	35 4 N ... 80 4 N	95 4 N ... 175 4 N	200 4 N ... 350 4 N	-
6-pole	7 6E ... 18 6 E	20 6	32 6 ... 50 6	32 6 N ... 60 6 N	80 6 N ... 165 6 N	190 6 N ... 480 6 N	530 6 N ... 850 6 N
8-pole	-	10 8 ... 17 8	26 8 ... 35 8	26 8 N ... 50 8 N	75 8 N ... 130 8 N	150 8 N ... 400 8 N	460 8 N ... 760 8 N
10-pole	-	-	-	-	40 10 N ... 90 10 N	110 10 N ... 350 10 N	390 10 N ... 660 10 N
12-pole	-	-	-	-	-	105 12 N ... 300 12 N	340 12 N ... 560 12 N
General	KSB standard (ZN 56525)						
Type of installation							
Stationary, with guide cable arrangement	Installation depth 15 ft [4.5 m] ²³⁾						
Stationary, with guide rail arrangement	Installation depth 15 ft [4.5 m] ²⁴⁾						
Transportable	Up to size 300-401, except sizes 200-500/501, 200-631, 250-630					-	
Stationary, dry	-			With cooling jacket			

Material variants H, C1, C2

Feature	Motors					
2-pole	3 2E ... 7 2E	5 2 ... 25 2	-	-	-	-
4-pole	2 4E ... 5 4E	4 4 ... 29 4	35 4 ... 65 4	80 4.N	95 4.N ... 175 4.N	200 4.N ... 350 4.N
6-pole	-	4 6 ... 19 6	32 6 ... 50 6	60 6.N	80 6.N ... 165 6.N	190 6.N ... 480 6.N
8-pole	-	-	26 8 ... 35 8	50 8.N	75 8.N ... 130 8.N	150 8.N ... 480 8.N
10-pole	-	-	-	-	-	110 10.N ... 350 10.N
12-pole	-	-	-	-	-	105 12.N ... 300 12.N
Suction flange	25)					
Shaft material for material variant H						
Shaft	A 276 Type 420			A 576 Gr. 1045	A 276 Type 420	
Shaft protecting sleeve	-	-	-	A 276 Type 420	A 276 Type 420	
Shaft material for material variant C1, C2						
Shaft	A 182 F51 / A 576 Gr. 1045			A 276 Type 420		
Shaft protecting sleeve	-	-	-	A 182 F51	A 182 F51	A 182 F51
Bearings	Grease-packed rolling element bearings sealed for life			Pump end: re-greasable rolling element bearings Drive end: grease-packed rolling element bearings sealed for life		
Explosion protection						
Versions U, W	Not explosion-proof					

22) Optional: S, D, E, F impellers to ISO 9906/A, K impellers to ISO 9906//1/2/A

23) Optional: up to 98 ft [30 m], from size 200-500 up to 49 ft [15 m]

24) Optional: up to 98 ft [30 m]

25) Drilled to ANSI B16.1 Class 125; not available for all sizes; on request

Feature	Motors					
2-pole	3 2E ... 7 2E	5 2 ... 25 2	-	-	-	-
4-pole	2 4E ... 5 4E	4 4 ... 29 4	35 4 ... 65 4	80 4.N	95 4.N ... 175 4.N	200 4.N ... 350 4.N
6-pole	-	4 6 ... 19 6	32 6 ... 50 6	60 6.N	80 6.N ... 165 6.N	190 6.N ... 480 6.N
8-pole	-	-	26 8 ... 35 8	50 8.N	75 8.N ... 130 8.N	150 8.N ... 480 8.N
10-pole	-	-	-	-	-	110 10.N ... 350 10.N
12-pole	-	-	-	-	-	105 12.N ... 300 12.N
Versions X, Z	 or  Explosion-proof Class I, Division I, Groups C & D, T3					-
Motor						
Starting method	DOL, soft starter, frequency inverter or star-delta					
Voltage	460 V ²⁶⁾					
Cooling	Cooled by surrounding fluid					
Immersion depth	100 ft [30 m] max.					
Power cable						
Type	See table "Overview of power cables"					
Length	30 ft [10 m] ²⁷⁾					
Cable entry	Absolutely watertight					
Sealing elements						
Elastomer seals	Nitrile butadiene rubber NBR ²⁸⁾ , for C2 generally fluorocarbon rubber FPM					
Shaft seal	C1: bellows-type mechanical seal ²⁹⁾ H, C2: mechanical seal with covered spring					Mechanical seal with covered spring
Monitoring equipment						
Winding temperature, version U, W	Temperature switch (bimetal) in the winding					
Winding temperature, version X, Z	Temperature switch (bimetal) in the winding, plus PTC for explosion protection					
Bearing temperature	-	-	- ³⁰⁾	Pt100 at the pump end ³¹⁾		
Leakage inside the motor	Leakage sensor in the motor space					
Mechanical seal leakage	-	-	Float switch in the leakage area for D impeller	Float switch in leakage area		
Vibration sensor	-	-	- ³²⁾			

²⁶⁾ Optional: 200 V, 230 V, 380 V, 575 V

²⁷⁾ Optional: up to 164 ft [50 m]

²⁸⁾ Optional: Viton = fluorocarbon rubber FPM

²⁹⁾ Optional: mechanical seal with covered spring

³⁰⁾ Optional: Pt100 at the pump end

³¹⁾ Optional: Pt100 at the drive end

³²⁾ Optional: internal vibration sensor

Feature	Motors					
2-pole	3 2E ... 7 2E	5 2 ... 25 2	-	-	-	-
4-pole	2 4E ... 5 4E	4 4 ... 29 4	35 4 ... 65 4	80 4.N	95 4.N ... 175 4.N	200 4.N ... 350 4.N
6-pole	-	4 6 ... 19 6	32 6 ... 50 6	60 6.N	80 6.N ... 165 6.N	190 6.N ... 480 6.N
8-pole	-	-	26 8 ... 35 8	50 8.N	75 8.N ... 130 8.N	150 8.N ... 480 8.N
10-pole	-	-	-	-	-	110 10.N ... 350 10.N
12-pole	-	-	-	-	-	105 12.N ... 300 12.N
Coating	Material variant H: environmentally friendly KSB standard coating, primer dry film thickness 1 1/2 - 2 mils (35-50 µm), top coat: two-component epoxy coating dry film thickness 6 mils (150 mm), color RAL 5002 ³³⁾ Material variants C1, C2: without coating					
Maximum fluid and ambient temperature						
Version U, X	104 °F [40 °C]					
Versions W, Z	140 °F [60 °C]					
Tests/inspections						
Hydraulic system	KSB standard (ZN 56525) ³⁴⁾					
General	KSB standard (ZN 56525)					
Type of installation						
Stationary, with guide cable arrangement	Installation depth 15 ft [4.5 m] ³⁵⁾					
Transportable	Installation depth 15 ft [4.5 m] ³⁶⁾					

Overview of power cables

Feature	S1BN8-F rubber-sheathed cable	S07RC4N8-F rubber-sheathed cable	TEHSITE Tefzel cable
Design	Standard	Optional	Optional
Rated voltage	1000 V	750 V	750 V
EMC screening	-	✓	-
Insulation material	EPR ³⁷⁾	EPR ³⁷⁾	ETFE ³⁸⁾
Max. continuous temperature of insulation	194 °F [90 °C]	194 °F [90 °C]	275 °F [135 °C]
For permanent immersion in waste water to DIN VDE 0282-16/HD22.16	✓	✓	✓

³³⁾ Optional: dry film thickness 12 mils (300 µm)

³⁴⁾ Optional: S, D, E, F impellers to Hydraulic Institute, Level B; K impellers to Hydraulic Institute, Level A

³⁵⁾ Optional: up to 98 ft [30 m], from size 200-500 up to 49 ft [15 m]

³⁶⁾ Up to size 300-401, except sizes 200-501, 200-631, 250-630

³⁷⁾ EPR = ethylene propylene rubber

³⁸⁾ ETFE = ethylene tetrafluoroethylene

Impellers

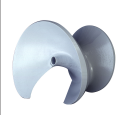
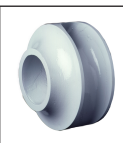
	Impeller with cutter (impeller type S/S-max)
	Free-flow impeller (impeller type F/F-max)
	Closed single-channel impeller (impeller type E/E-max)
	Open, diagonal single-vane impeller (impeller type D)
	Closed multi-channel impeller (impeller type K/K-max)

Table of fluids handled

The table below for your guidance is based on KSB's long-standing experience. The data are standard values and are not to be considered as generally binding recommendations. More detailed advice is available from KSB. Make use of our laboratory's expertise when selecting materials.

Selection aid for materials and hydraulic systems per fluid

Fluid handled ³⁹⁾	Recommended material	Recommended impeller type ⁴⁰⁾	Comments, further recommendations
Gray water	Gray cast iron	K/K-max, D, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
River water	Gray cast iron	K/K-max, D, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
Stormwater	Gray cast iron	K/K-max, D, E/E-max, F/F-max	Free passage > any solids contained, possibly pre-screened
Waste water:			
▪ Untreated municipal waste water	Gray cast iron	F/F-max, S/S-max, D, E/E-max, K/K-max	ATV ⁴¹⁾ recommends a free passage of 100 mm, min. 76 mm
▪ Containing air and gas	Gray cast iron	F/F-max	Up to 8 %, contact KSB for fluids with high outgassing rates
Sludges:			
▪ Raw sludge	Gray cast iron	F/F-max, D, E/E-max	Pumpable up to a dry substance content of: 13 % (D), 8 % (F), 6 % (E)
▪ Digested sludge	Gray cast iron	F/F-max, D, E/E-max	Pumpable up to a dry substance content of: 13 % (D), 8 % (F), 6 % (E)
▪ Activated sludge	Gray cast iron	D, K/K-max	Pumpable up to a dry substance content of: 13 % (D), 5 % (K)
Industrial waste water containing:			
▪ Paint suspensions	Gray cast iron	K/K-max	Solvent-free, observe the operator's instructions.
▪ Lacquer/paint/varnish suspensions	Gray cast iron	F/F-max, E/E-max	Solvent-free, contact KSB for silicone-free version
▪ Fibers/pulp	Gray cast iron	F/F-max, S/S-max, D	-
▪ Chips/swarf	Gray cast iron	K/K-max, F/F-max	Material variants G2 or GH, special mechanical seal, solids content < 5 g/l
▪ Abrasive substances ⁴²⁾	Gray cast iron	K/K-max, F/F-max	Material variants G2 or GH, special mechanical seal, solids content < 5 g/l
Mildly acidic industrial waste water	Gray cast iron	K/K-max, F/F-max	pH ≥ 6.5, material variant C1 and FPM (Viton) O-rings
Neutral non-corrosive waste water:			
▪ Ammonium hydroxide	Gray cast iron	K/K-max	-
▪ Ammonium hydroxide 5 % NH ₄ OH	Gray cast iron	K/K-max	-
▪ Urea 25 % (NH ₂) ₂ -CO	Gray cast iron	K/K-max	-
▪ Potassium hydroxide 10 % KOH	Gray cast iron	K/K-max	-
▪ Calcium hydroxide 5 % Ca(OH) ₂	Gray cast iron	K/K-max	-
▪ Sodium hydroxide 5 % NaOH	Gray cast iron	K/K-max	-
▪ Sodium carbonate 30 % Na ₂ CO ₃	Gray cast iron	K/K-max	-

³⁹⁾ For any fluids which are not listed in this table contact KSB.

⁴⁰⁾ The first impeller type listed should be given preference.

⁴¹⁾ ATV = German regulatory body for waste water management

⁴²⁾ Severe hydroabrasive wear occurs if solids contents of 0.5 g/l or higher are combined with circumferential speeds exceeding 20 m/s or part-load conditions to the left of the duty point.

Fluid handled ³⁹⁾	Recommended material	Recommended impeller type ⁴⁰⁾	Comments, further recommendations
Neutral, non-corrosive waste water containing:			
▪ Aliphatic hydrocarbons, e.g. oils, petrol, butane, methane	Gray cast iron	K/K-max	-
▪ Aromatic hydrocarbons, e.g. benzene, styrene	Gray cast iron	K/K-max	FPM (Viton) O-rings ⁴³⁾
▪ Chlorinated hydrocarbons, e.g. tetrachloroethylene, ethylene chloride, chloroform, methylene chloride	Gray cast iron	K/K-max	FPM (Viton) O-rings ⁴³⁾
Highly abrasive industrial waste water causing wear (chemically neutral): ⁴⁴⁾			
▪ Water containing iron ore sinter	Wear-resistant white cast iron	K/K-max	Sinter content < 5 g/l: material variant GH Sinter content > 5 g/l: material variant H
▪ Lime milk containing quartz and pigment suspensions	Wear-resistant white cast iron	K/K-max	Lime milk content < 15 %: material variant GH Lime milk content > 15 %: material variant H
▪ Wash water containing solids	Wear-resistant white cast iron	K/K-max, F/F-max	Material selection based on fluid analysis
▪ Waste water containing dust or ash	Wear-resistant white cast iron	K/K-max	Material selection based on fluid analysis
Wasser/sand mixture	Wear-resistant white cast iron	K/K-max, F/F-max	Solids content < 5 g/l: material variant GH Solids content > 5 g/l: material variant H
Seawater	Duplex stainless steel	K/K-max, F/F-max	Material variant C2 ≤ 25 °C (fluid temperature) ⁴⁵⁾
Brackish water	Duplex stainless steel	K/K-max, F/F-max	Material variant C1 or G1 (with 250 µm two-component epoxy resin), depending on salt content
Corrosive industrial waste water	Duplex stainless steel	K/K-max, F/F-max	Material variant C1 or C2 based on fluid analysis

³⁹⁾ For any fluids which are not listed in this table contact KSB.

⁴⁰⁾ The first impeller type listed should be given preference.

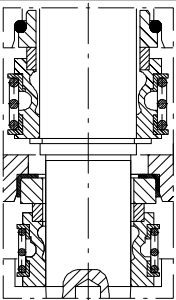
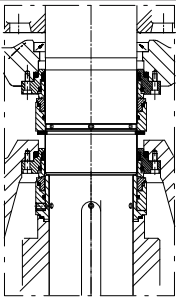
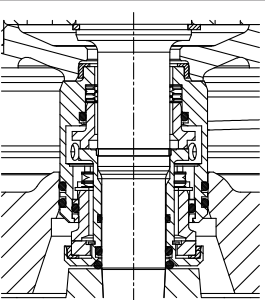
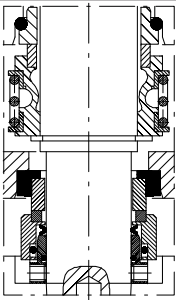
⁴³⁾ The hydrocarbons mentioned may occur in very high concentrations due to the difference in specific weight and their low solubility. If this is the case, contact KSB.

⁴⁴⁾ The required material variants highly depend on the operating hours, speed and flow velocity.

⁴⁵⁾ Higher fluid temperatures on request.

Shaft seal

Available shaft seal types per bearing bracket

Standard design		Standard variant ⁴⁶⁾	
Mechanical seal with elastomer bellows (NBR, optional: Viton) ⁴⁷⁾	Stationary cartridge seal with spring outside of fluid	Cartridge seal ⁴⁸⁾⁴⁹⁾⁵⁰⁾	Product-side mechanical seal with covered spring ⁴⁹⁾⁵⁰⁾
			

⁴⁶⁾ A surcharge and longer delivery times apply to standard variants.

⁴⁷⁾ For all types of waste water

⁴⁸⁾ Only possible in combination with motor generation "E" for maximum nominal impeller diameters 215, 216, 217

⁴⁹⁾ For very abrasive fluids or fluids containing metallic particles (e.g. shavings from drilling)

⁵⁰⁾ Standard for material variants H and C2 (optional for material variants G, G1, G2, GH and C1)

Technical data

Gray cast iron (G, G1, G2, GH)

Size		Material variant	Impeller				Dry installation (installation types D, H)		Wet installation (installation types S, P, K)		Mass moment of inertia J ₅₁
			Number of impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	Max. operating pressure ⁵²⁾	Max. test pressure	
S 40-252	G	4	9/32	9 1/4	6 7/8	-	-	145,0	188,6	0,03	
S 50-216	G	4	3/16	8 1/4	6 7/8	-	-	130,5	174,0	0,018	
F 40-252	G, G1, G2, GH	-	1	8 1/4	5 7/8	-	-	110,0	142,1	0,03	
F 50-215	G, G1, G2, GH	-	1 5/8	8 1/4	5 1/8	-	-	116,0	152,3	0,019	
F 50-216	G, G1, G2, GH	-	1	8 1/4	4 11/16	-	-	94,3	123,3	0,025	
F 65-215	G, G1, G2, GH	-	2 9/16	8 1/4	4 11/16	-	-	79,8	108,8	0,025	
F 65-217	G, G1, G2, GH	-	2 9/16	7 7/8	4 3/4	101,0	152,0	87,0	116,0	0,020	
F 80-216	G, G1, G2, GH	-	3	8 1/4	4 11/16	-	-	72,5	94,3	0,025	
F 80-217	G, G1, G2, GH	-	3	7 7/8	4 3/4	87,0	131,0	73,0	95,0	0,022	
F 80-252	G, G1, G2, GH	-	3	10 7/16	5 7/8	87,0	130,5	91,4	118,9	0,14	
F 80-253	G, G1, G2, GH	-	3	10 7/16	5 7/8	87,0	130,5	91,4	118,9	0,14	
F 100-215	G, G1, G2, GH	-	3 15/16	8 1/4	4 11/16	-	-	58,0	79,8	0,025	
F 100-254	G, G1, G2, GH	-	4	10 7/16	7 7/8	87,0	130,5	49,3	65,2	0,056	
F 100-316	G, G1, G2, GH	-	4	11 7/16	9 5/16	-	-	71,1	92,8	0,075	
F 100-401	G, G1, G2, GH	-	4	15 3/8	12 13/16	145,0	217,5	110,2	142,1	0,248	
F 150-315	G, G1, G2, GH	-	4 3/4	11 7/16	9 13/16	87,0	130,5	26,1	33,4	0,144	
F 150-401	G, G1, G2, GH	-	5 5/16	15 3/8	10 5/8	145,0	217,5	61,0	79,8	0,248	
E 65-216	G	1	2 9/16	7 1/16	5 1/2	-	-	80,0	104,0	0,020	
E 80-253	G	1	3	10 5/8	8 7/8	87,0	130,5	59,5	78,3	0,17	
E 100-253	G	1	3 1/8 - 3 3/4	9 15/16	7 1/4	87,0	130,5	49,3	65,3	0,18	
E 100-255	G	1	3 9/16	9 15/16	7 1/4	65,0	98,0	51,0	73,0	0,160	
E 100-317	G	1	3	12 15/16	11 1/4	101,0	152,0	87,0	116,0	0,250	
E 150-317	G	1	4 1/2 - 4 3/4	12 5/8	10	87,0	130,5	45,0	59,5	0,31	
E 150-401	G	1	4 1/2 - 5 1/2	16	13 11/16	145,0	217,5	91,4	118,9	0,68	
E 200-401	G	1	4 3/4 - 5 1/2	15 3/4	12 9/16	145,0	217,5	82,7	107,3	0,86	
D 80-315	G, G1	1	2 1/2	10 1/4	9 1/16	145,0	217,5	150,8	197,3	0,124	
D 100-253	G, G1	1	3	10 7/16	9 3/16	87,0	130,5	50,8	66,7	0,115	
D 100-316	G, G1	1	3 3/8	12 1/16	10 5/8	-	-	52,2	68,2	0,233	
D 150-253	G, G1	1	4	10	8 7/8	87,0	130,5	27,5	34,8	0,115	
D 150-315	G, G1	1	4	12 1/2	11	87,0	130,5	47,9	62,4	0,289	
D 150-400	G, G1	1	4	14 15/16	12 13/16	-	-	75,4	98,6	0,573	
D 150-401	G, G1	1	4 3/8	15 1/8	14 9/16	-	-	76,9	100,1	0,999	
D 200-315	G, G1	1	4	12 3/8	11	87,0	130,5	39,2	49,3	0,261	
D 200-400	G, G1	1	4	14 3/4	14	-	-	60,9	79,8	0,825	
D 250-400	G, G1	1	4 3/4	14 9/16	12 9/16	-	-	50,8	66,7	0,653	
D 300-400	G, G1	1	6	16 1/16	14 3/4	-	-	24,7	31,9	0,925	
K 40-252	G, G1, GH	3	5/8	10 1/4	5 7/8	-	-	145,0	188,5	0,047	
K 80-253	G, G1, GH	2	1 1/8	8 11/16	5 1/2	-	-	95,7	124,7	0,15	
K 100-255	G, G1, G2, GH	2	3	9 7/8	6 7/8	-	-	52,0	68,0	0,07	
K 100-316	G, G1, GH	2	3	11 5/8	9 1/4	-	-	71,1	92,8	0,13	
K 100-400	G, G1	2	3	16 1/16	14	-	-	133,4	174,1	1,1	

⁵¹⁾ Data applies to maximum impeller diameter and impeller with water fill

⁵²⁾ Permissible operating pressure = inlet pressure + pressure at Q = 0

Size	Material variant	Impeller				Dry installation (installation types D, H)		Wet installation (installation types S, P, K)		Mass moment of inertia J ₅₁
		Number of impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	Max. operating pressure ⁵²⁾	Max. test pressure	
K 100-401	G, G1, GH	2	2	15 7/8	12 3/16	145,0	217,5	134,9	175,5	0,504
K 150-315	G, G1, GH	2	3	12 3/16	9 1/4	87,0	130,5	50,8	66,7	0,18
K 150-317	G, G1, G2, GH	2	3	12 3/16	9 7/8	87,0	131,0	73,0	95,0	0,280
K 150-400	G	3	3	15 7/8	11 13/16	-	-	130,5	169,6	0,83
K 150-401	G, G1, GH	2	3	15 7/8	12 3/16	145,0	217,5	129,1	168,2	0,916
K 200-315	G, G1, GH	3	2 3/4	11 5/8	9 5/8	87,0	130,5	27,6	34,8	0,22
K 200-316	G, G1, GH	2	4	12	10 7/16	87,0	130,5	24,7	31,9	0,22
K 200-317	G, G1, G2, GH	3	3	12 3/16	9 5/8	57,0	87,0	44,0	57,0	0,40
K 200-318	G, G1, G2, GH	2	3 15/16	12 3/16	9 1/16	57,0	87,0	44,0	57,0	0,28
K 200-330	G, G1, GH	3	2 3/4	12 13/16	11 5/16	145,0	217,5	75,4	100,1	0,35
K 200-400	G, G1	3	3	16 1/16	11 13/16	-	-	98,6	127,6	0,52
K 200-401	G, G1, GH	3	3 1/8	15 7/8	13	145,0	217,5	103,0	133,4	0,52
K 200-501	G, G1	2	4 1/8	19 3/4	17 11/16	145,0	217,5	92,8	120,4	1,68
K 200-631	G, G1	2	4 1/8	24 1/2	21 1/4	145,0	217,5	142,1	185,6	4,41
K 250-400	G, G1, GH	3	3 3/8	14 9/16	11 13/16	145,0	217,5	95,7	123,3	0,5
K 250-401	G, G1, GH	2	4 1/8	15 3/4	12 3/16	145,0	217,5	87,0	113,1	0,55
K 250-630	G, G1	4	3 1/2	24 13/16	19 11/16	145,0	217,5	150,8	195,8	2,76
K 250-900	G, G1	3	4 5/16	22 1/16	28 1/4	232,0	348,0	169,7	220,5	19,03
K 300-400	G, G1	3	4	16 1/16	13 1/16	145,0	217,5	50,8	66,7	0,75
K 300-401	G, G1, GH	2	5 5/16	16 1/16	14 7/16	145,0	217,5	33,4	42,1	0,75
K 300-420	G, G1	3	4	16 1/16	14 9/16	87,0	130,5	50,8	105,9	0,95
K 300-500	G, G1	3	3 1/2	19 13/16	16 15/16	145,0	217,5	90,0	116,0	1,48
K 300-503	G, G1	5	2	18 7/8	15 15/16	145,0	217,5	129,1	168,2	2,5
K 350-420	G, G1	3	4	17 11/16	15 1/4	87,0	130,5	50,8	66,7	1,22
K 350-500	G, G1	3	4 5/16	20	16 3/4	87,0	130,5	82,7	107,3	3,12
K 350-501	G, G1	2	6 5/8	20 1/16	19 1/2	87,0	130,5	40,6	53,6	3
K 350-630	G, G1	3	5 5/16	24 13/16	19 11/16	145,0	217,5	105,9	136,3	5,22
K 350-636	G, G1	5	3	23 7/16	20 1/16	145,0	217,5	92,8	120,4	5,42
K 350-710	G, G1	3	4 1/4	28 3/4	22 13/16	145,0	217,5	136,3	176,9	10,6
K 400-500	G, G1	3	5 1/8	20	17 7/16	87,0	130,5	49,3	65,2	3,37
K 400-630	G, G1	3	5 1/4	24 7/16	21 1/2	87,0	130,5	89,9	116,0	8,21
K 400-710	G, G1	3	6 1/2	29 1/8	23 1/8	145,0	217,5	127,6	166,8	14
K 400-900	G, G1	3	5 1/8	32 11/16	25 15/16	232,0	348,0	163,9	213,2	17,79
K 500-630	G, G1	3	5 1/4	22 15/16	20 7/16	58,0	87,0	60,9	79,8	6,11
K 500-632	G, G1	3	5 1/4	22 15/16	20 7/16	80,0	124,0	65,0	86,0	9,36
K 500-710	G, G1	3	6	29 1/8	23 1/8	916	174,1	100,1	130,5	13,0
K 500-900	G, G1	3	7 15/16	35 3/4	28 3/8	145,0	217,5	116,0	149,4	45
K 600-520	G, G1	3	5 11/16	20 15/16	18	58,0	87,0	34,8	46,4	7,02
K 600-710	G, G1	3	6 1/2	29	26 15/16	58,0	87,0	60,9	79,8	16,96
K 700-900	G, G1	3	7 1/2	33 7/16	29 1/16	43,5	65,0	47,9	62,4	40
K 700-901	G, G1	3	7 1/16	35 3/4	29 15/16	130,5	195,8	104,4	134,9	50

⁵¹⁾ Data applies to maximum impeller diameter and impeller with water fill

⁵²⁾ Permissible operating pressure = inlet pressure + pressure at Q = 0

Gray cast iron (G, G1, G2, GH)

Size		Material variant	Impeller			Dry installation (installation types D, H)		Wet installation (installation types S, P, K)		Mass moment of inertia j ₅₃	
			Number of impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	Max. operating pressure ⁵²⁾		Max. test pressure
S 40-252	G	4	7	235	175	-	-	10	13	0,03	
S 50-216	G	4	4	210	175	-	-	9	12	0,018	
F 40-252	G, G1, G2, GH	-	25	210	150	-	-	7,6	9,8	0,03	
F 50-215	G, G1, G2, GH	-	42	210	130	-	-	8	10,5	0,019	
F 50-216	G, G1, G2, GH	-	25	210	120	-	-	6,5	8,5	0,025	
F 65-215	G, G1, G2, GH	-	65	210	120	-	-	5,5	7,5	0,025	
F 65-217	G, G1, G2, GH	-	65	200	120	7	10,5	6,0	8,0	0,020	
F 80-216	G, G1, G2, GH	-	76	210	120	-	-	5,0	6,5	0,025	
F 80-217	G, G1, G2, GH	-	76	200	120	6	9	5,0	6,5	0,022	
F 80-252	G, G1, G2, GH	-	76	265	150	6	9	6,3	8,2	0,14	
F 80-253	G, G1, G2, GH	-	76	265	150	6	9	6,3	8,2	0,14	
F100-215	G, G1, G2, GH	-	100	210	120	-	-	4,0	5,5	0,025	
F 100-254	G, G1, G2, GH	-	100	265	200	6	9	3,4	4,5	0,056	
F 100-316	G, G1, G2, GH	-	100	290	236	-	-	4,9	6,4	0,075	
F 100-401	G, G1, G2, GH	-	100	390	325	10	15	7,6	9,8	0,248	
F 150-315	G, G1, G2, GH	-	120	290	250	6	9	1,8	2,3	0,144	
F 150-401	G, G1, G2, GH	-	135	390	270	10	15	4,2	5,5	0,248	
E 65-216	G	1	65	180	140	-	-	5,5	7,5	0,20	
E 80-216	G	1	76	210	160	-	-	5,5	7,5	0,035	
E 80-253	G	1	76	270	225	6	9	4,1	5,4	0,17	
E 100-253	G	1	80-95	252	184	6	9	3,4	4,5	0,18	
E 100-255	G	1	90	252	184	4,5	7	3,5	5,0	0,16	
E 100-317	G	1	76	328	286	7,0	10,5	6,0	8,0	0,25	
E 150-317	G	1	110-120	320	254	6	9	3,1	4,1	0,31	
E 150-401	G	1	115-140	407	348	10	15	6,3	8,2	0,68	
E 200-401	G	1	121-143	400	319	10	15	5,7	7,4	0,86	
D 80-315	G, G1	1	65	260	230	10	15	10,4	13,6	0,124	
D 100-253	G, G1	1	76	265	234	6	9	3,5	4,6	0,115	
D 100-315	G, G1	1	75	222	196	-	-	6,8	8,8	0,065	
D 100-316	G, G1	1	85	306	270	-	-	3,6	4,7	0,233	
D 150-253	G, G1	1	100	254	225	6	9	1,9	2,4	0,115	
D 150-315	G, G1	1	100	317	280	6	9	3,3	4,3	0,289	
D 150-400	G, G1	1	100	363	326	-	-	5,2	6,8	0,573	
D 150-401	G, G1	1	110	384	370	-	-	5,3	6,9	0,999	
D 200-315	G, G1	1	100	315	280	6	9	2,7	3,4	0,261	
D 200-400	G, G1	1	100	375	355	-	-	4,2	5,5	0,825	
D 250-400	G, G1	1	120	370	320	-	-	3,5	4,6	0,653	
D 300-400	G, G1	1	150	408	375	-	-	1,7	2,2	0,925	
K 40-252	G, G1, GH	3	15	260	150	-	-	10	13	0,047	
K 80-253	G, G1, GH	2	33	220	140	-	-	6,6	8,6	0,15	
K 100-255	G, G1, G2, GH	2	76	250	175	-	-	3,6	4,7	0,07	
K 100-316	G, G1, GH	2	76	295	235	-	-	4,9	6,4	0,13	
K 100-400	G, G1	2	76	408	355	-	-	9,2	12	1,1	
K 100-401	G, G1, GH	2	50	404	310	10	15	9,3	12,1	0,504	
K 150-315	G, G1, GH	2	76	310	235	6	9	5,0	6,5	0,28	
K 150-317	G, G1, G2, GH	2	76	309	250	6	9	3,5	4,6	0,18	
K 150-400	G	3	76	404	300	-	-	9,0	11,7	0,83	
K 150-401	G, G1, GH	2	76	404	310	10	15	8,9	11,6	0,916	

⁵³⁾ Data applies to maximum impeller diameter and impeller with water fill

Size		Material variant	Impeller				Dry installation (installation types D, H)		Wet installation (installation types S, P, K)		Mass moment of inertia j ⁵³⁾
			Number of impeller channels	Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	Max. operating pressure ⁵²⁾	Max. test pressure	
K 200-315	G, G1, GH	3	70	295	245	6	9	1,9	2,4	0,22	
K 200-316	G, G1, GH	2	100	305	265	6	9	1,7	2,2	0,22	
K 200-317	G, G1, G2, GH	3	76	309	244	4	6	3,0	4,0	0,40	
K 200-318	G, G1, G2, GH	2	100	309	230	4	6	3,0	4,0	0,28	
K 200-330	G, G1, GH	3	70	326	287	10	15	5,2	6,8	0,35	
K 200-400	G, G1	3	76	408	300	-	-	6,8	8,8	0,52	
K 200-401	G, G1, GH	3	80	404	330	10	15	7,1	9,2	0,52	
K 200-501	G, G1	2	105	502	450	10	15	6,4	8,3	1,68	
K 200-631	G, G1	2	105	622	540	10	15	9,8	12,8	4,41	
K 250-400	G, G1, GH	3	85	370	300	10	15	6,6	8,5	0,5	
K 250-401	G, G1, GH	2	105	400	310	10	15	6	7,8	0,55	
K 250-630	G, G1	4	90	630	500	10	15	10,4	13,5	2,76	
K 250-900	G, G1	3	110	840	717	16	24	11,7	15,2	19,03	
K 300-400	G, G1	3	100	408	332	10	15	3,5	4,6	0,75	
K 300-401	G, G1, GH	2	135	408	367	10	15	2,3	2,9	0,75	
K 300-420	G, G1	3	100	408	370	6	9	5,6	7,3	0,95	
K 300-500	G, G1	3	90	504	430	10	15	6,2	8	1,48	
K 300-503	G, G1	5	50	480	405	10	15	8,9	11,6	2,5	
K 350-420	G, G1	3	100	450	387	6	9	3,5	4,6	1,22	
K 350-500	G, G1	3	110	508	426	6	9	5,7	7,4	3,12	
K 350-501	G, G1	2	170	509	495	6	9	2,8	3,7	3	
K 350-630	G, G1	3	135	630	500	10	15	7,3	9,4	5,22	
K 350-636	G, G1	5	75	595	510	10	15	6,4	8,3	5,42	
K 350-710	G, G1	3	110	730	580	10	15	9,4	12,2	10,6	
K 400-500	G, G1	3	130	508	443	6	9	3,4	4,5	3,37	
K 400-630	G, G1	3	132	620	546	6	9	6,2	8	8,21	
K 400-710	G, G1	3	165	739	587	10	15	8,8	11,5	14	
K 400-900	G, G1	3	130	830	659	16	24	11,3	14,7	17,79	
K 500-630	G, G1	3	133	582	520	4	6	4,2	5,5	6,11	
K 500-632	G, G1	3	135	639	560	5,5	8,5	4,5	5,9	9,36	
K 500-710	G, G1	3	150	739	586	8	12	6,9	9,0	13,0	
K 500-900	G, G1	3	202	908	721	10	15	8	10,3	45	
K 600-520	G, G1	3	145	532	457	4	6	2,4	3,2	7,02	
K 600-710	G, G1	3	165	736	685	4	6	4,2	5,5	16,96	
K 700-900	G, G1	3	190	850	738	3	4,5	3,3	4,3	40	
K 700-901	G, G1	3	180	908	760	9	13,5	7,2	9,3	50	

⁵³⁾ Data applies to maximum impeller diameter and impeller with water fill

Industrial materials (H, C1, C2)

Size	Material variant	Number of impeller channels	Impeller			Wet installation (installation type S, P)		Mass moment of inertia [J51]
			Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	
F 40-252	H, C1, C2	-	1	8 1/4	5 7/8	114,6	142,1	0,03
F 50-215	H, C1, C2	-	1 5/8	8 1/4	5 1/8	116,0	152,3	0,019
F 50-216	H, C1, C2	-	1	8 1/4	4 11/16	94,3	123,3	0,025
F 65-215	H, C1, C2	-	2 9/16	8 1/4	4 11/16	79,8	108,8	0,025
F 80-216	H, C1, C2	-	3	8 1/4	4 11/16	72,5	94,3	0,025
F 100-254	H, C1, C2	-	4	10 7/16	7 7/8	49,3	65,3	0,056
F 100-316	H, C1, C2	-	4	11 7/16	9 5/16	71,1	92,8	0,075
F 100-401	H, C1, C2	-	4	15 3/8	12 13/16	110,2	142,1	0,248
F 150-315	H, C1, C2	-	4 3/4	11 7/16	9 13/16	26,1	33,4	0,144
F 150-401	H, C1, C2	-	5 5/16	15 3/8	10 5/8	60,9	79,8	0,248
K 40-252	H, C1, C2	3	5/8	10 1/4	5 7/8	145,0	188,6	0,047
K 80-253	H, C1, C2	2	1 1/8	8 11/16	5 1/2	95,0	124,7	0,15
K 100-255	H, C1, C2	2	3	9 7/8	6 7/8	52,0	68,02	0,07
K 100-316	H, C1, C2	2	3	11 5/8	9 1/4	71,1	92,8	0,13
K 100-400	C1, C2	3	3	16 1/16	14	137,8	179,9	1,1
K 100-401	H, C1, C2	2	2	15 7/8	12 3/16	134,9	175,5	0,504
K 150-315	H, C1, C2	2	3	12 3/16	9 1/4	50,8	66,7	0,18
K 150-317	H, C1, C2	2	3	12 3/16	9 7/8	73,0	95,0	0,28
K 150-401	H, C1, C2	2	3	15 7/8	12 3/16	129,1	168,2	0,916
K 200-315	H, C1, C2	3	2 3/4	11 5/8	9 5/8	27,6	34,8	0,22
K 200-316	H, C1, C2	2	4	12	10 7/16	24,7	31,9	0,22
K 200-330	H, C1, C2	3	2 3/4	12 13/16	11 5/16	75,4	98,6	0,35
K 200-401	H, C1, C2	3	3 1/8	15 7/8	13	103,0	98,6	0,52
K 200-501	C1, C2	2	4 1/8	19 3/4	17 11/16	92,8	120,4	1,68
K 200-631	C1, C2	2	4 1/8	24 1/2	21 1/4	142,1	185,6	4,41
K 250-400	H, C1, C2	3	3 3/8	14 9/16	11 13/16	95,7	123,3	0,5
K 250-401	H, C1, C2	2	4 1/8	15 3/4	12 3/16	87,0	113,1	0,55
K 250-630	C1, C2	3	3 9/16	24 13/16	19 11/16	150,8	195,8	2,76
K 300-400	H, C1, C2	3	4	16 1/16	13 1/16	50,8	66,7	0,75
K 300-401	H, C1, C2	2	5 5/16	16 1/16	14 7/16	33,4	42,1	0,75
K 300-420	C1, C2	3	4	16 1/16	14 9/16	81,2	105,9	0,95
K 300-500	C1, C2	3	3 9/16	19 13/16	16 15/16	90,0	116,0	1,48
K 300-503	C1, C2	5	2	18 7/8	15 15/16	129,1	168,2	2,5
K 350-420	C1, C2	3	4	17 11/16	15 1/4	50,8	66,7	1,22
K 350-500	C1, C2	3	4 5/16	20	16 3/4	82,7	107,3	3,12
K 350-630	C1, C2	3	5 5/16	24 13/16	19 11/16	105,9	136,3	5,22
K 350-636	C1, C2	5	3	23 7/16	20 1/16	92,8	120,4	5,42
K 350-710	C1, C2	3	4 5/16	28 3/4	22 13/16	136,3	177,0	10,6
K 400-500	C1, C2	3	5 1/8	20	17 7/16	49,3	65,3	3,37
K 400-630	C1, C2	3	5 1/4	24 7/16	21 1/2	90,0	116,0	8,21
K 500-630	C1, C2	3	5 1/4	22 15/16	20 7/8	60,9	79,8	6,11
K 600-520	C1, C2	3	5 3/4	20 15/16	18	34,8	46,4	7,02
K 600-710	C1, C2	3	6 1/2	29	26 15/16	60,9	79,8	16,96
K 700-900	C1, C2	3	7 1/2	33 7/16	29 1/16	47,9	62,4	40
K 700-901	C1, C2	3	7 1/16	35 3/4	29 15/16	104,4	134,9	50

Industrial materials (H, C1, C2)

Size	Material variant	Number of impeller channels	Impeller			Wet installation (installation types S, P)		Mass moment of inertia J ₅₁
			Free passage	Max. impeller diameter	Min. impeller diameter	Max. operating pressure ⁵²⁾	Max. test pressure	
F 40-252	H, C1, C2	-	25	210	150	7,6	9,8	0,03
F 50-215	H, C1, C2	-	42	210	130	8,0	10,5	0,019
F 50-216	H, C1, C2	-	25	210	120	6,5	8,5	0,025
F 65-215	H, C1, C2	-	65	210	120	5,5	7,5	0,025
F 80-216	H, C1, C2	-	76	210	120	5,0	6,5	0,025
F 100-254	H, C1, C2	-	100	265	200	3,4	4,5	0,056
F 100-316	H, C1, C2	-	100	290	236	4,9	6,4	0,075
F 100-401	H, C1, C2	-	100	390	325	7,6	9,8	0,248
F 150-315	H, C1, C2	-	120	290	250	1,8	2,3	0,144
F 150-401	H, C1, C2	-	135	390	270	4,2	5,5	0,248
K 40-252	H, C1, C2	3	15	260	150	10	13	0,047
K 80-253	H, C1, C2	2	33	220	140	6,6	8,6	0,15
K 100-255	H, C1, C2	2	76	250	175	3,6	4,7	0,07
K 100-316	H, C1, C2	2	76	295	235	4,9	6,4	0,13
K 100-400	C1, C2	3	76	408	355	9,5	12,4	1,1
K 100-401	H, C1, C2	2	50	404	310	9,3	12,1	0,504
K 150-315	H, C1, C2	2	76	310	235	3,5	4,6	0,18
K 150-317	H, C1, C2	2	76	309	250	5,0	6,5	0,28
K 150-401	H, C1, C2	2	76	404	310	8,9	11,6	0,916
K 200-315	H, C1, C2	3	70	295	245	1,9	2,4	0,22
K 200-316	H, C1, C2	2	100	305	265	1,7	2,2	0,22
K 200-330	H, C1, C2	3	70	326	287	5,2	6,8	0,35
K 200-401	H, C1, C2	3	80	404	330	7,1	9,2	0,52
K 200-501	C1, C2	2	105	502	450	6,4	8,3	1,68
K 200-631	C1, C2	2	105	622	540	9,8	12,8	4,41
K 250-400	H, C1, C2	3	85	370	300	6,6	8,5	0,5
K 250-401	H, C1, C2	2	105	400	310	6	7,8	0,55
K 250-630	C1, C2	3	90	630	500	10,4	13,5	2,76
K 300-400	H, C1, C2	3	100	408	332	3,5	4,6	0,75
K 300-401	H, C1, C2	2	135	408	367	2,3	2,9	0,75
K 300-420	C1, C2	3	100	408	370	5,6	7,3	0,95
K 300-500	C1, C2	3	90	504	430	6,2	8	1,48
K 300-503	C1, C2	5	50	480	405	8,9	11,6	2,5
K 350-420	C1, C2	3	100	450	387	3,5	4,6	1,22
K 350-500	C1, C2	3	110	508	426	5,7	7,4	3,12
K 350-630	C1, C2	3	135	630	500	7,3	9,4	5,22
K 350-636	C1, C2	5	75	595	510	6,4	8,3	5,42
K 350-710	C1, C2	3	110	730	580	9,4	12,2	10,6
K 400-500	C1, C2	3	130	508	443	3,4	4,5	3,37
K 400-630	C1, C2	3	132	620	546	6,2	8	8,21
K 500-630	C1, C2	3	133	582	520	4,2	5,5	6,11
K 600-520	C1, C2	3	145	532	457	2,4	3,2	7,02
K 600-710	C1, C2	3	165	736	685	4,2	5,5	16,96
K 700-900	C1, C2	3	190	850	738	3,3	4,3	40
K 700-901	C1, C2	3	180	908	760	7,2	9,3	50

Mass moments of inertia per motor
2-pole

Motor	Motor type	J
		[kgm²]
3 2 E	1	0,002
4 2 E	1	0,005
5 2 E	1	0,005
7 2 E	1	0,006
11 2 E	2	0,017
15 2 E	2	0,020
18 2 E	2	0,039
22 2 E	2	0,046
26 2 E	2	0,054
37 2	3	0,130
55 2	3	0,140

4-pole

Motor	Motor type	J
		[kgm²]
2 4 E	1	0,003
3 4 E	1	0,004
4 4 E	1	0,010
5 4 E	1	0,012
7 4 E	2	0,028
11 4 E	2	0,033
15 4 E	2	0,058
18 4 E	2	0,068
22 4 E	2	0,082
23 4	3	0,070
29 4	3	0,110
35 4	3	0,220
50 4	3	0,250
65 4	3	0,300
35 4 N	4	0,250
50 4 N	4	0,280
65 4 N	4	0,330
80 4 N	4	0,460
95 4 N	4	0,550
110 4 N	4	0,630
130 4 N	4	1,260
155 4 N	4	1,430
175 4 N	4	1,570
200 4 N	4	3,780
250 4 N	4	4,130
300 4 N	4	4,820
350 4 N	4	5,510

6-pole

Motor	Motor type	J
		[kgm²]
7 6 E	2	0,037
11 6 E	2	0,043
15 6 E	2	0,097
18 6 E	2	0,120
20 6	3	0,100
26 6	3	0,130
32 6	3	0,340
40 6	3	0,420
50 6	3	0,510
32 6 N	4	0,370
40 6 N	4	0,450
50 6 N	4	0,540
60 6 N	4	0,660
80 6 N	4	0,800
100 6 N	4	0,940

Motor	Motor type	J
		[kgm²]
120 6 N	4	1,890
140 6 N	4	2,250
165 6 N	4	2,550
190 6 N	4	7,300
225 6 N	4	8,570
260 6 N	4	9,840
320 6 N	4	14,30
360 6 N	4	15,90
400 6 N	4	17,60
440 6 N	4	19,20
480 6 N	4	20,70
530 6 N	4	31,50
580 6 N	4	36,30
630 6 N	4	41,10
690 6 N	4	45,80
770 6 N	4	50,60
850 6 N	4	55,30

8-pole

Motor	Motor type	J
		[kgm²]
10 8	3	0,090
17 8	3	0,120
21 8	3	0,180
26 8	3	0,370
35 8	3	0,470
26 8 N	4	0,400
35 8 N	4	0,500
50 8 N	4	0,660
75 8 N	4	0,940
90 8 N	4	1,980
110 8 N	4	2,250
130 8 N	4	2,550
150 8 N	4	7,300
185 8 N	4	8,570
220 8 N	4	9,840
260 8 N	4	13,30
300 8 N	4	15,90
350 8 N	4	19,10
400 8 N	4	20,70
460 8 N	4	31,50
530 8 N	4	36,30
580 8 N	4	41,10
630 8 N	4	45,80
690 8 N	4	50,60
760 8 N	4	55,30

10-pole

Motor	Motor type	J
		[kgm²]
40 10 N	4	1,750
60 10 N	4	1,930
75 10 N	4	2,200
90 10 N	4	2,490
110 10 N	4	7,960
150 10 N	4	9,660
190 10 N	4	11,80
230 10 N	4	17,70
270 10 N	4	20,50
310 10 N	4	23,20
350 10 N	4	25,80
390 10 N	4	36,10
430 10 N	4	41,60
475 10 N	4	47,20

Motor	Motor type	J
		[kgm²]
535 10 N	4	52,70
600 10 N	4	58,20
660 10 N	4	63,70

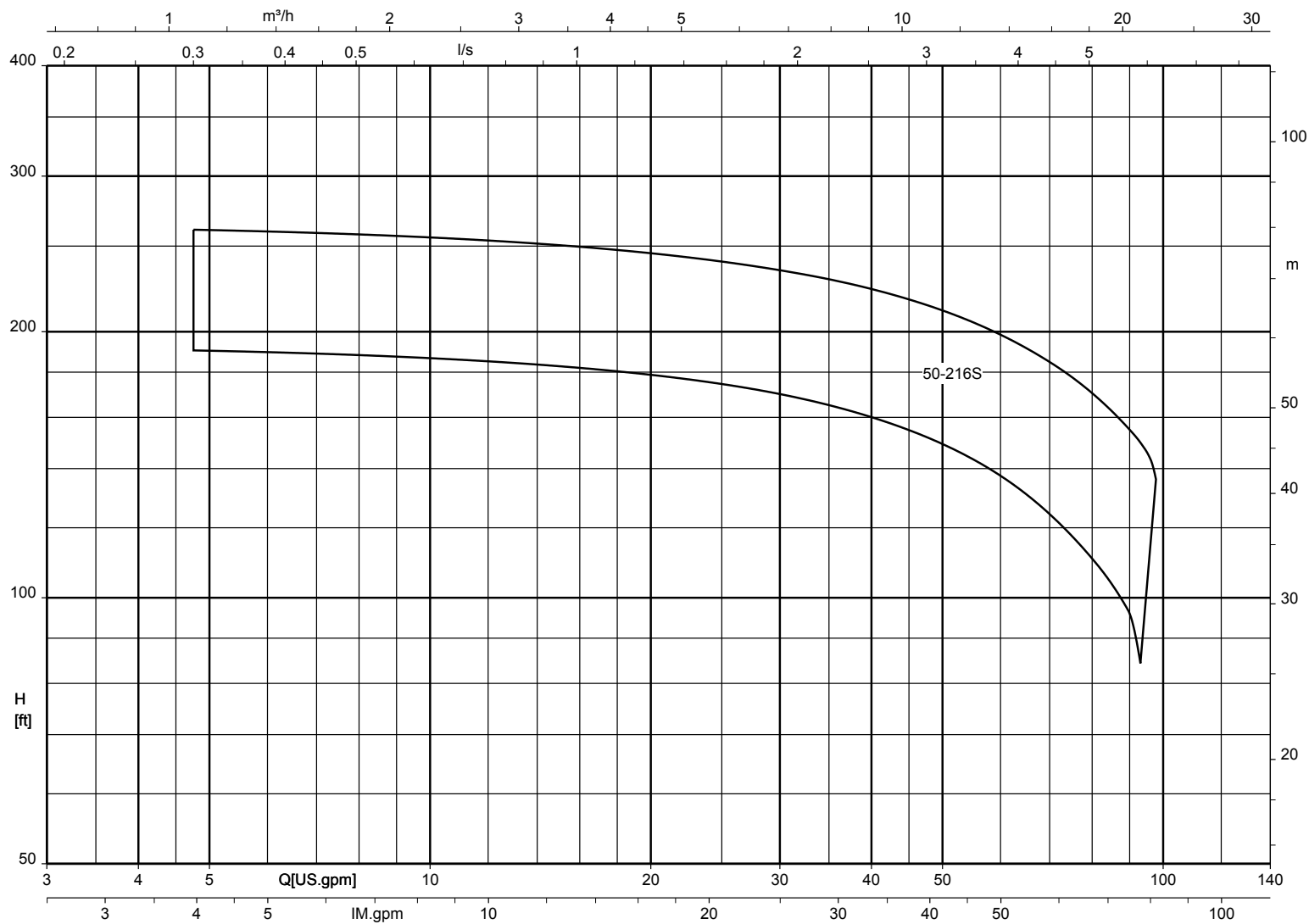
12-pole

Motor	Motor type	J
		[kgm²]
105 12 N	4	7,960
135 12 N	4	9,660
165 12 N	4	11,80

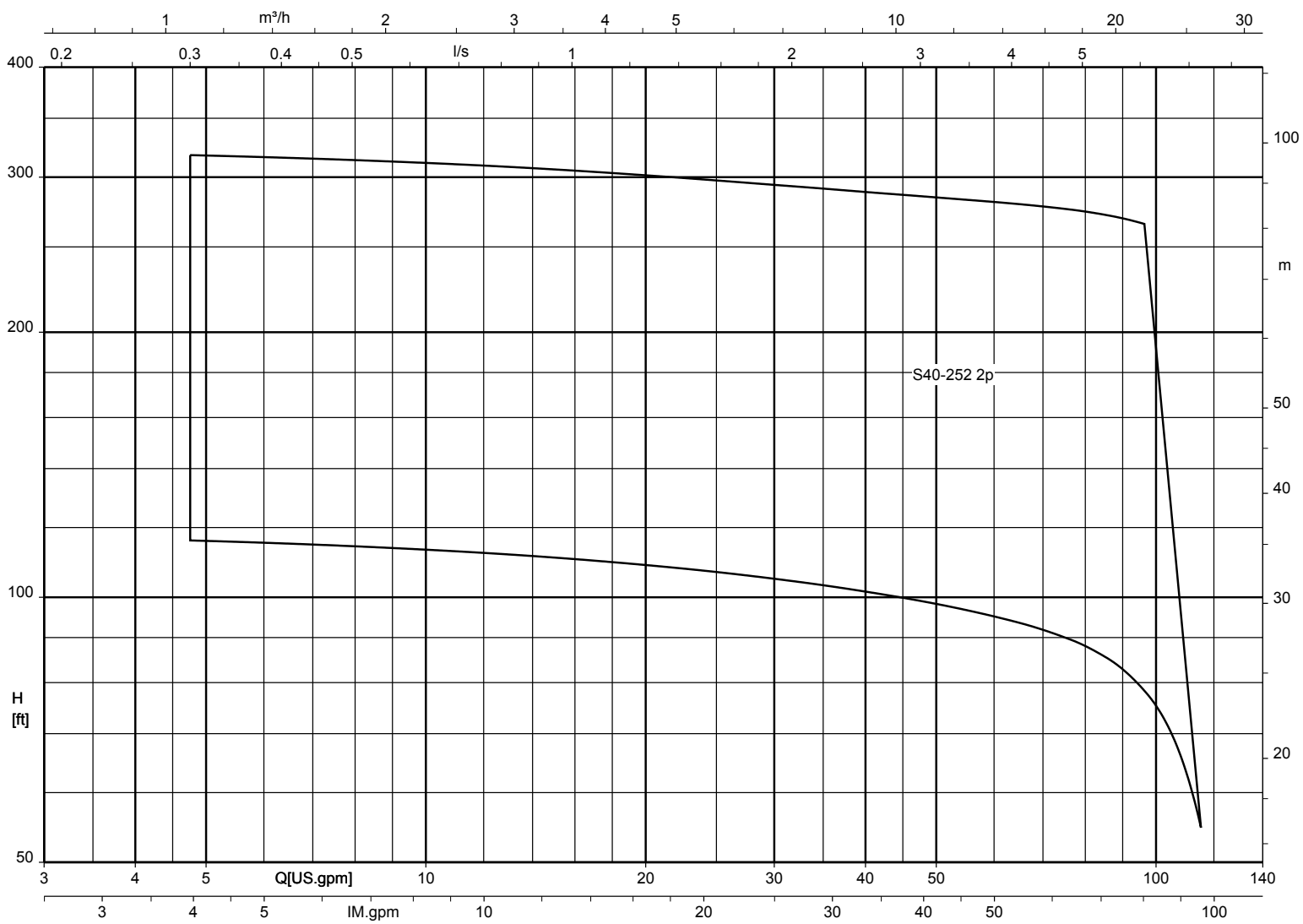
Motor	Motor type	J
		[kgm²]
195 12 N	4	17,70
230 12 N	4	20,50
265 12 N	4	23,20
290 12 N	4	36,10
300 12 N	4	25,80
340 12 N	4	41,60
380 12 N	4	47,20
450 12 N	4	52,70
490 12 N	4	58,20
560 12 N	4	63,70

Selection charts

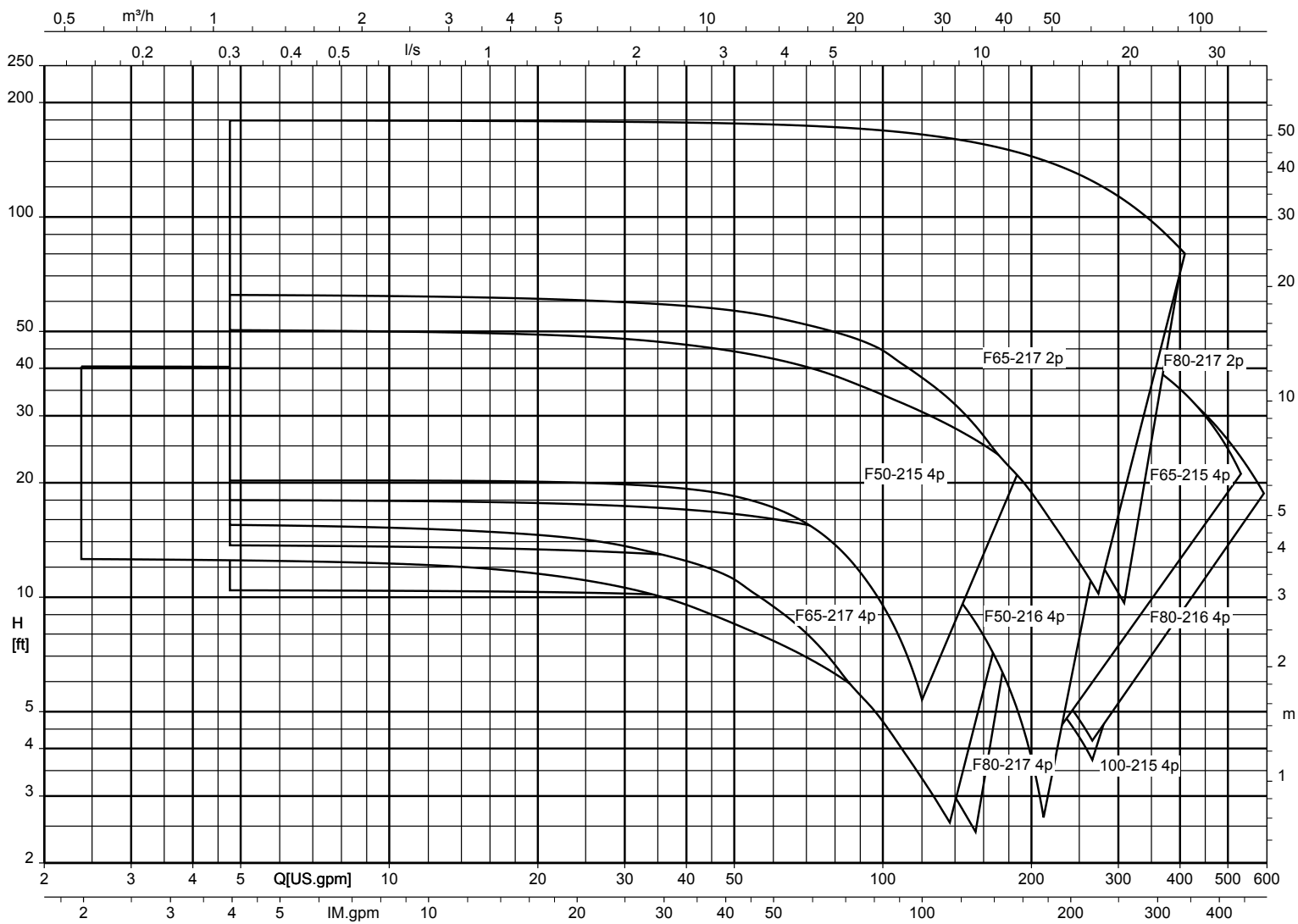
Amarex KRT S-max, n = 3500 rpm



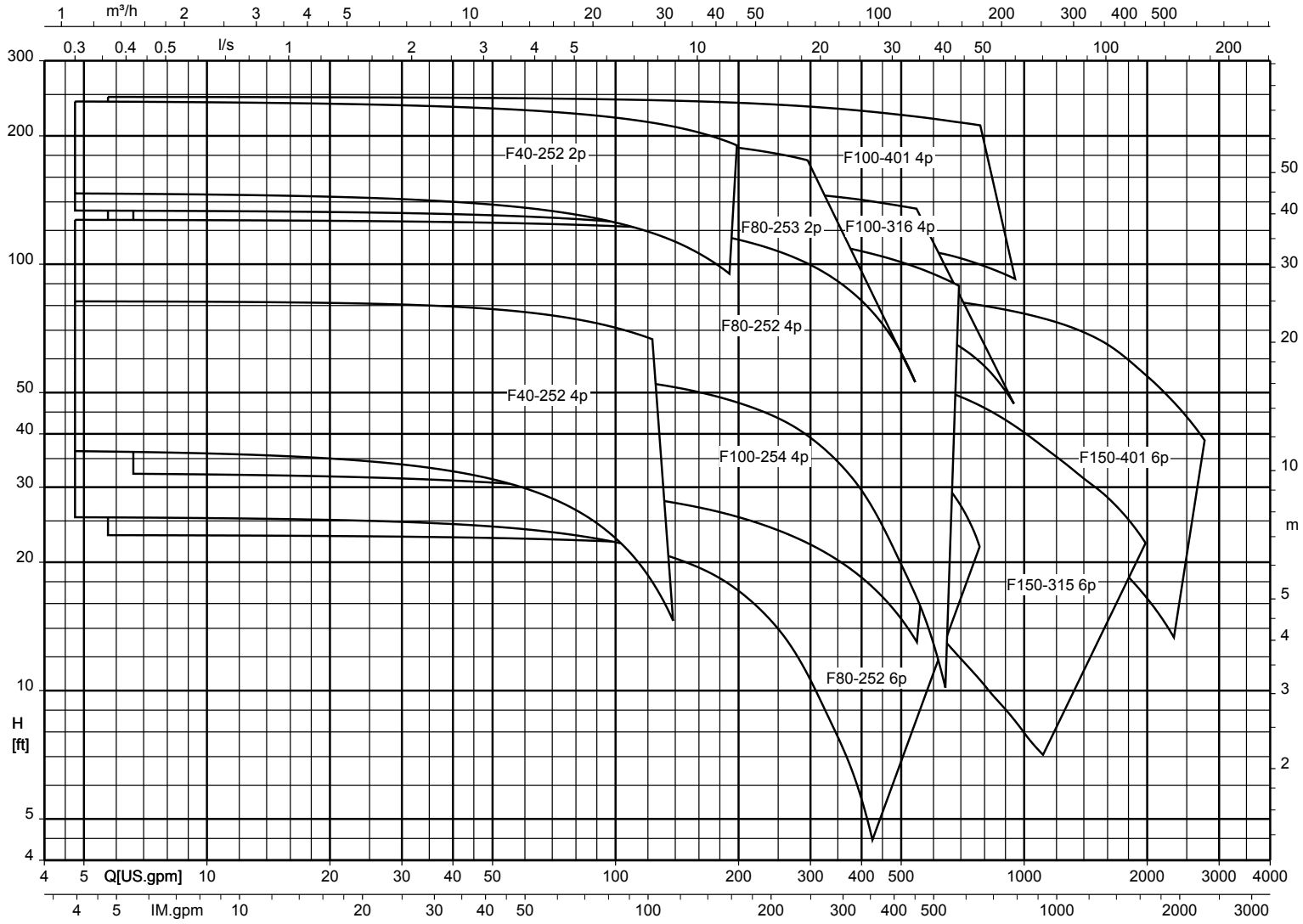
Amarex KRT S, n = 3500 rpm



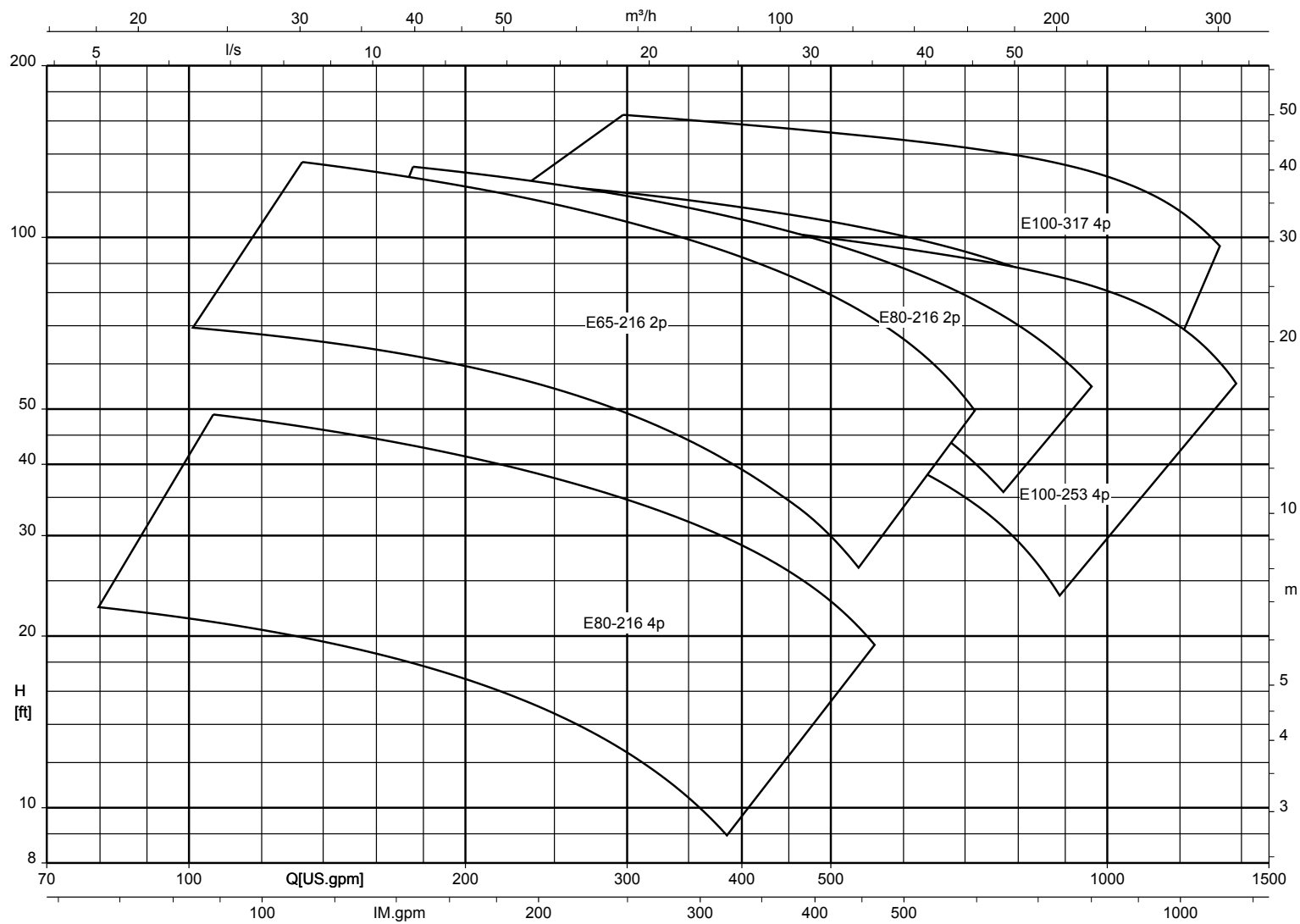
Amarex KRT F-max, n = 3500/1750 rpm



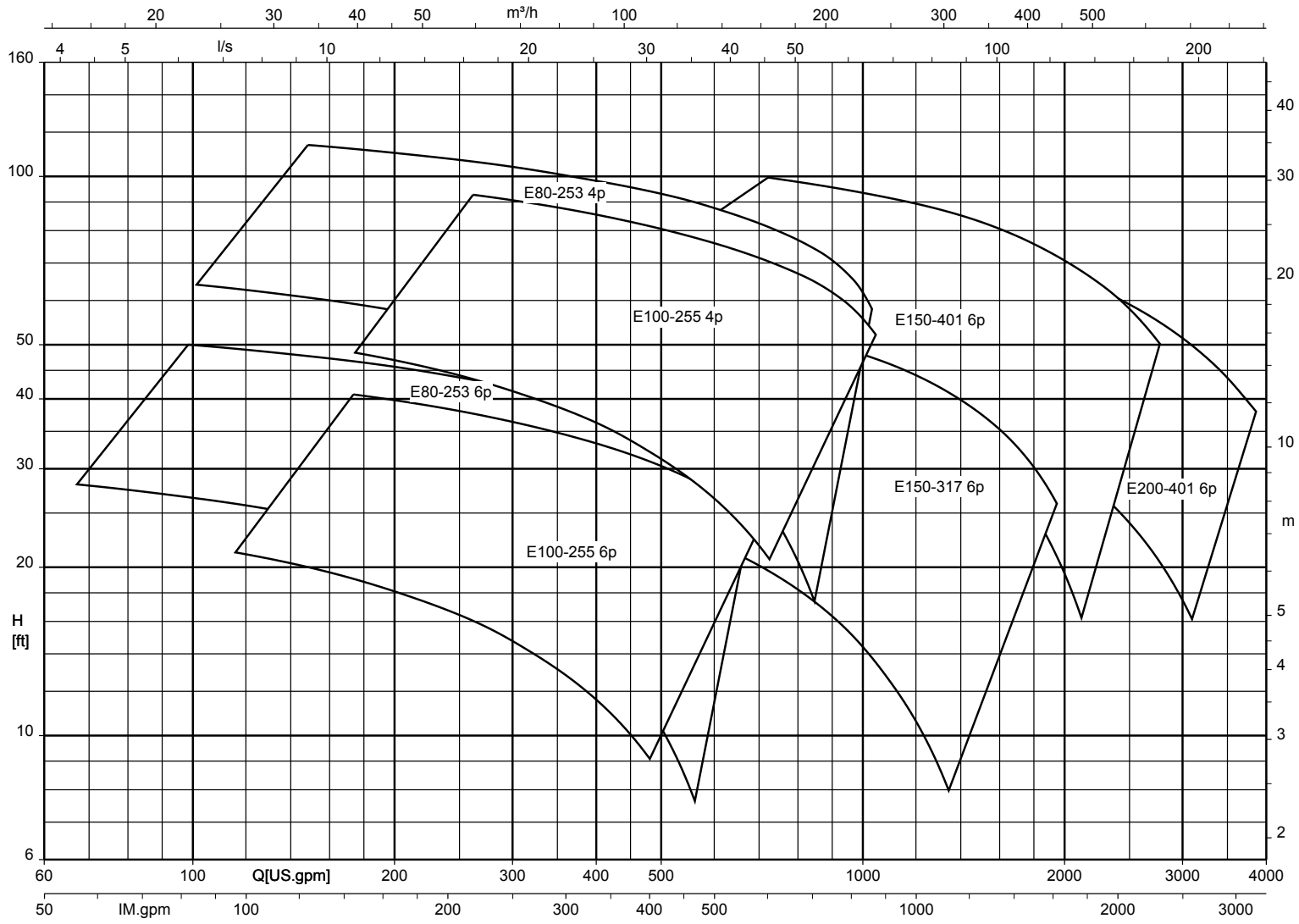
Amarex KRT F, n = 3500/1750/1160 rpm



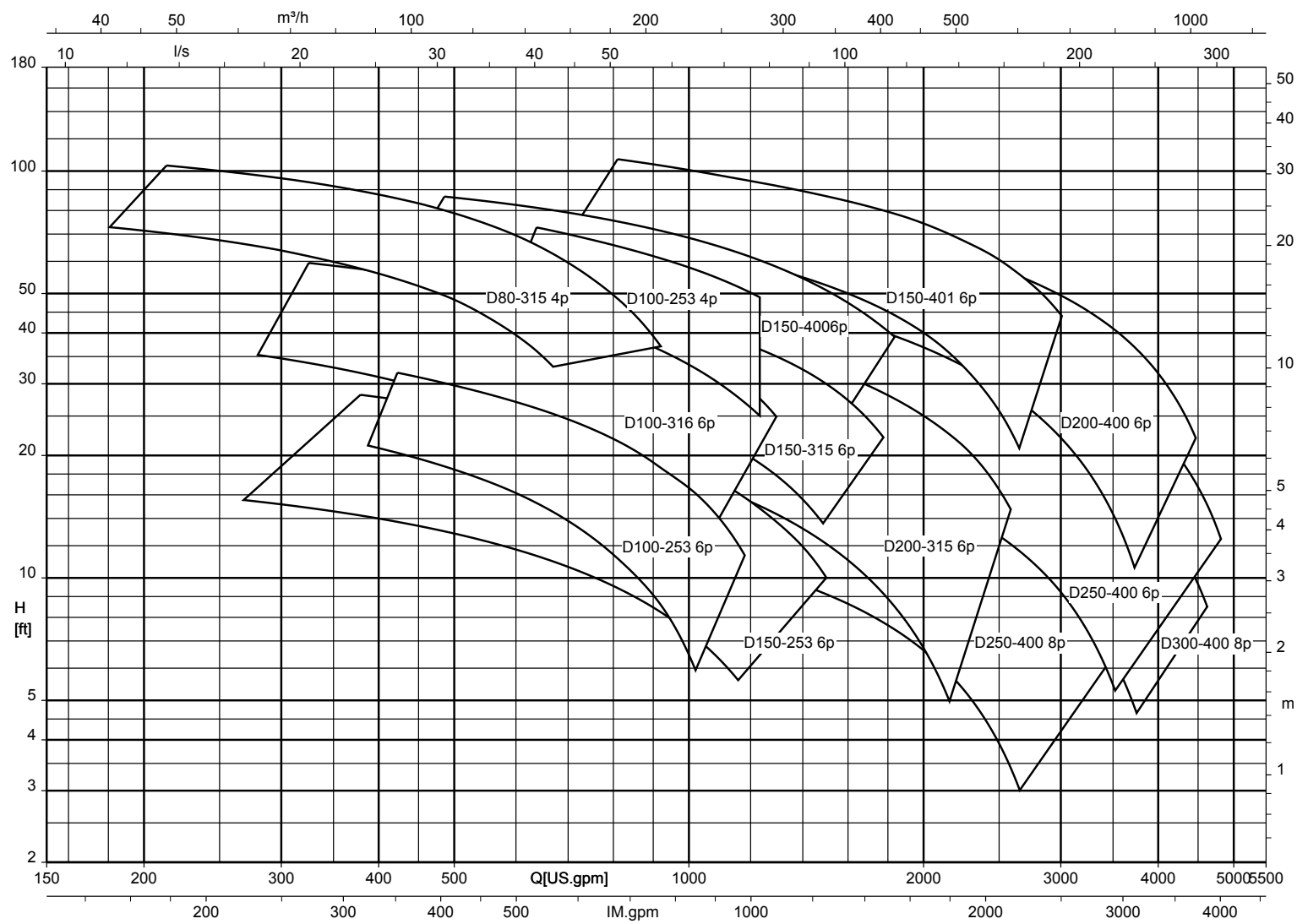
Amarex KRT E-max, n = 3500/1750 rpm



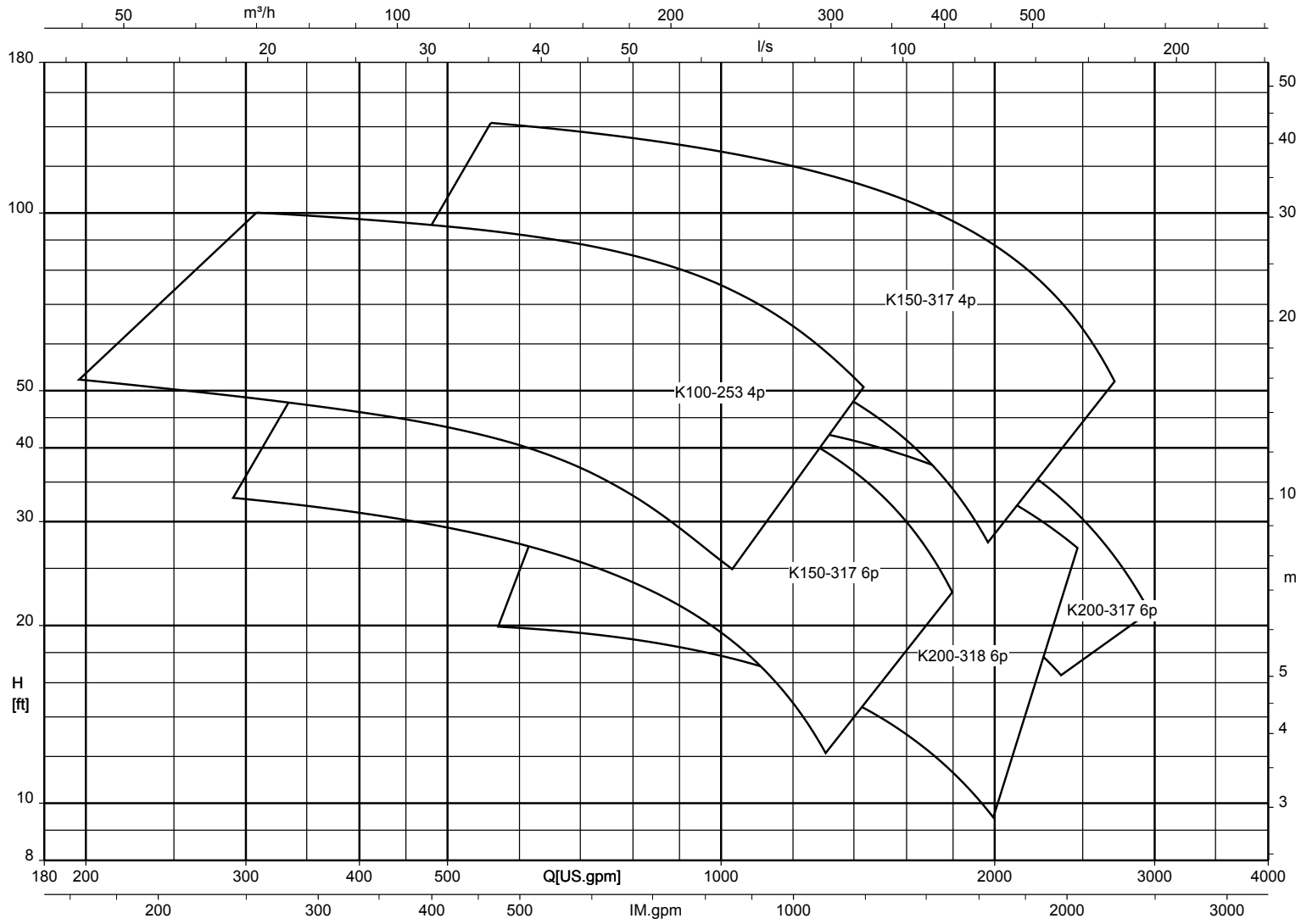
Amarex KRT E, n = 1750/1160 rpm



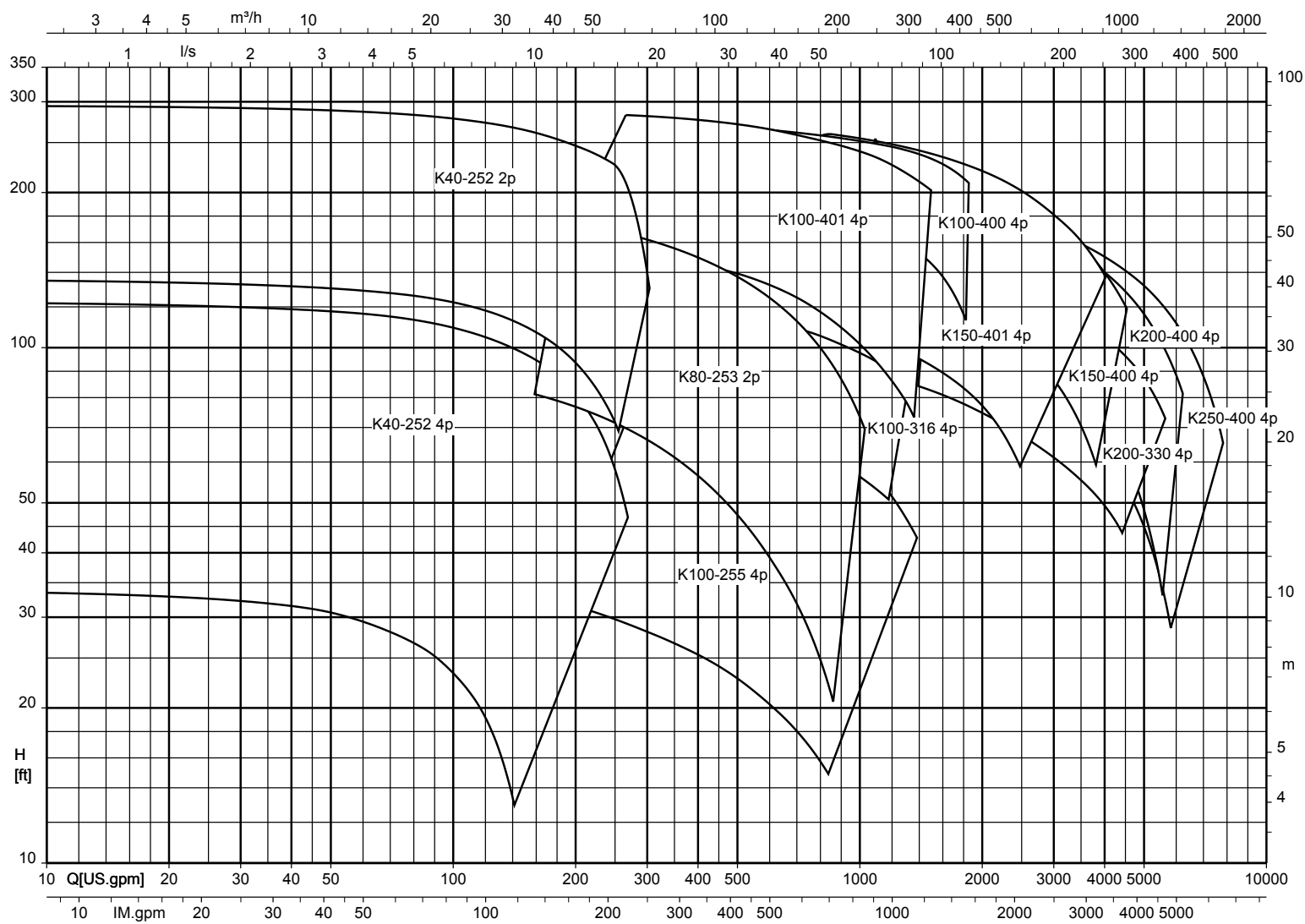
Amarex KRT D, n = 1750/1160/875 rpm



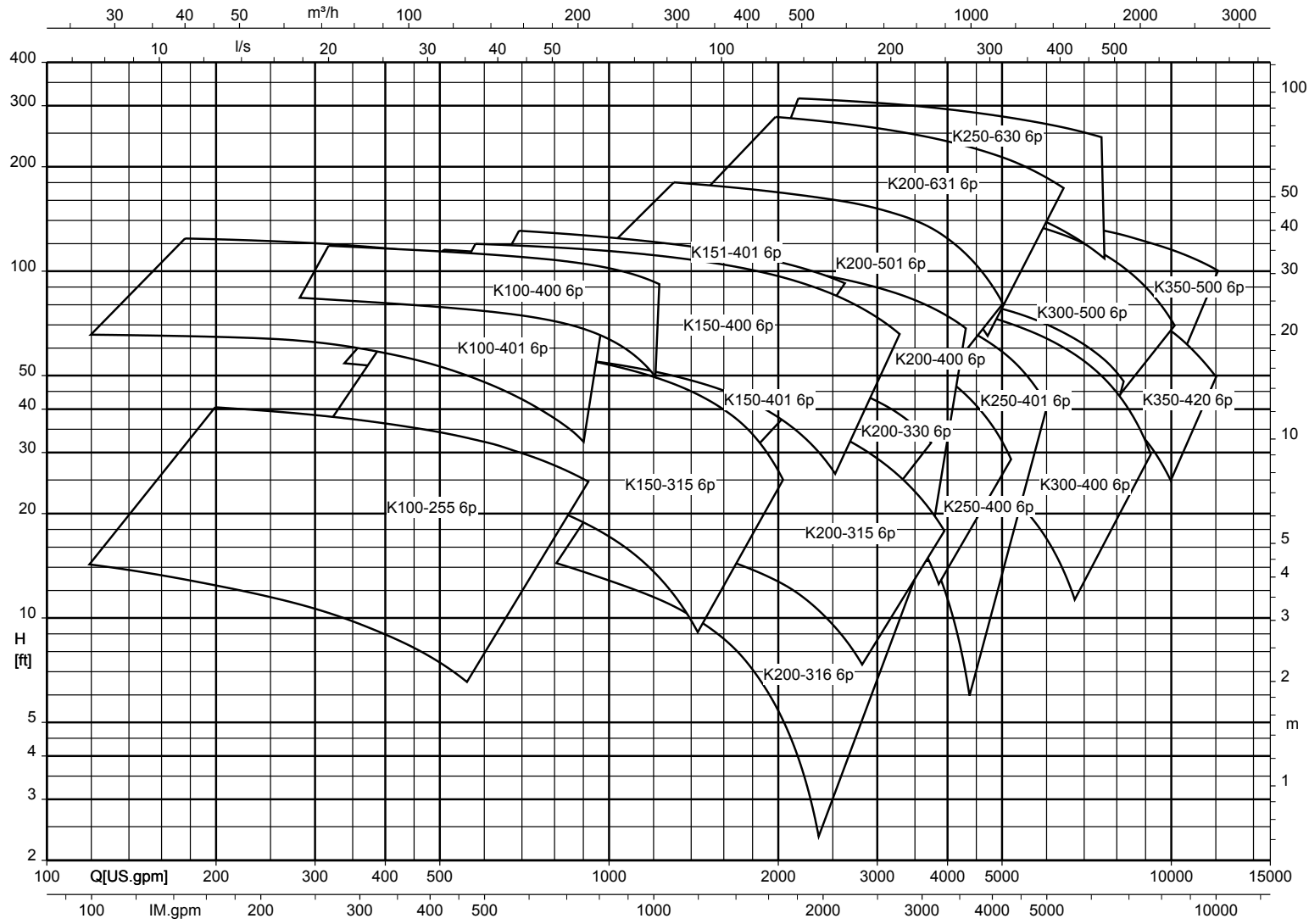
Amarex KRT K-max, n = 1750/1160 rpm



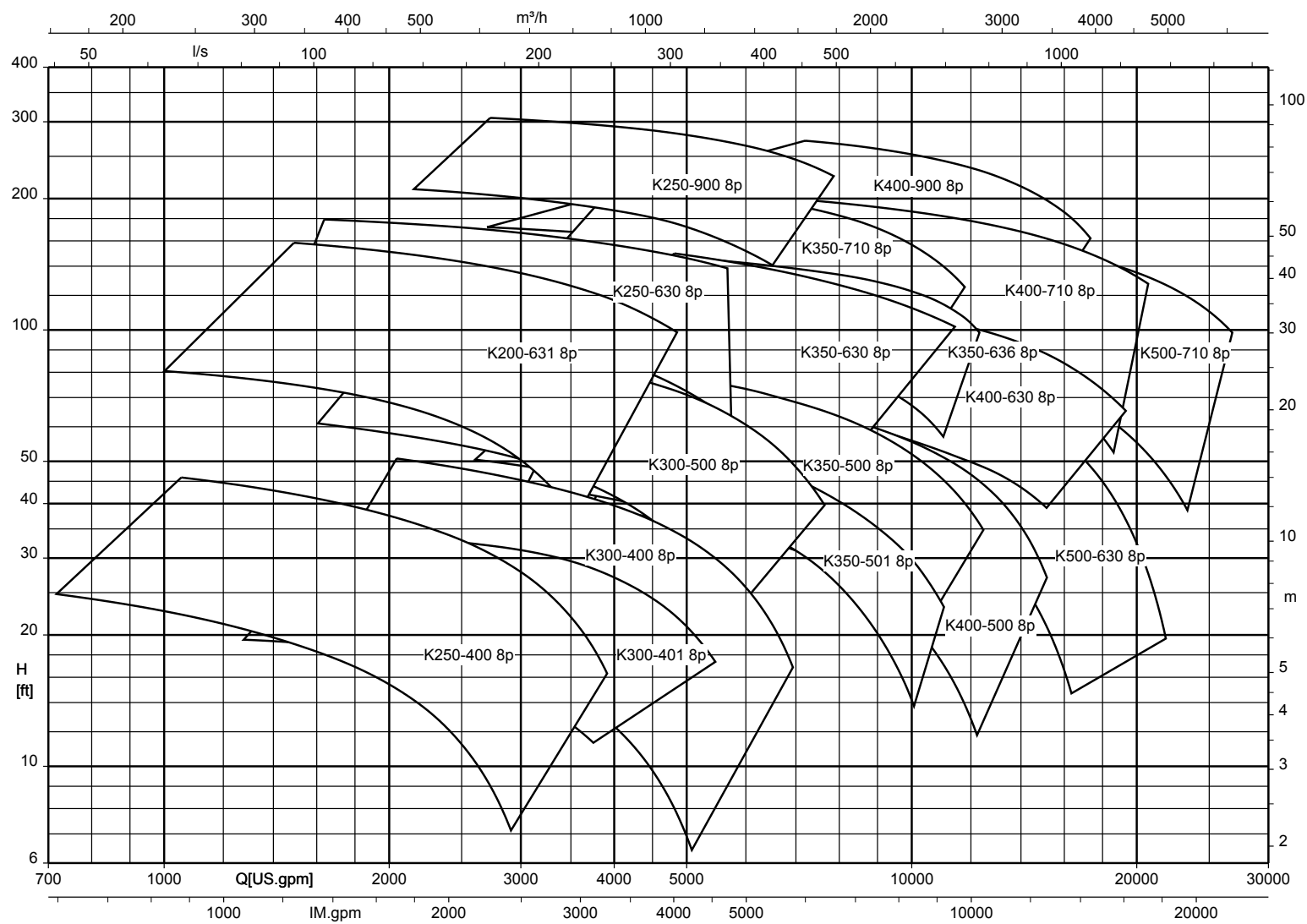
Amarex KRT K 40-.../80-.../100-.../150-.../200-.../250-..., n = 3500/1750 rpm



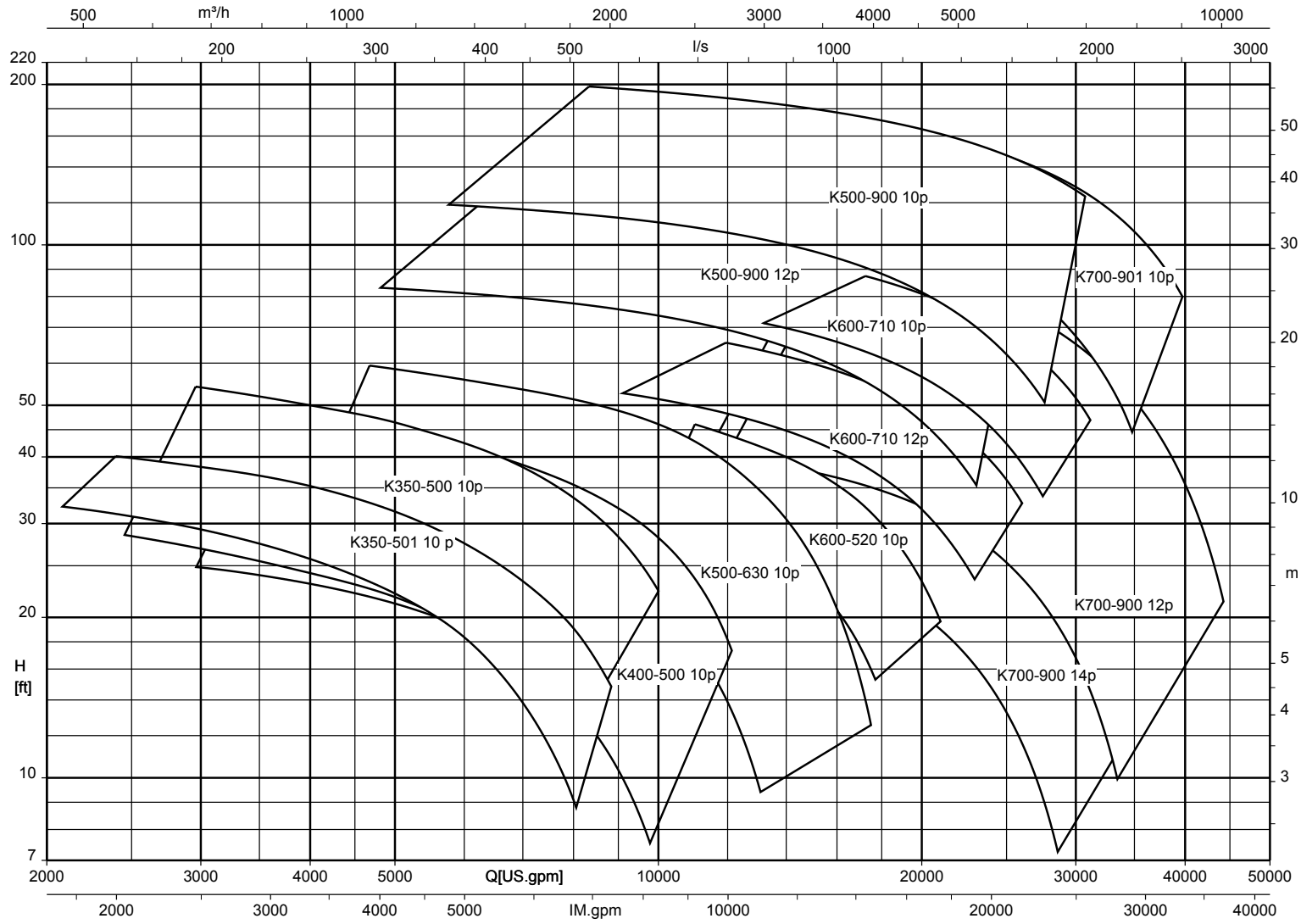
Amarex KRT K 100-.../150-.../151-.../200-.../250-.../300-.../350-..., n = 1160 rpm



Amarex KRT K 200-.../250-.../300-.../350-.../400-.../500-..., n = 875 rpm



Amarex KRT K 350-.../500-.../600-.../700-..., n = 700/585/500 rpm



Installation types

- Installation type D: stationary dry installation, vertical (S1 operation)
- Installation type H: stationary dry installation, horizontal (S1 operation)
- Installation type K: stationary wet installation (suitable for S1 operation with motor outside of the fluid)
- Installation type S: stationary wet installation (S1 operation with submerged motor)
- Installation type P: transportable wet installation (S1 operation with submerged motor)

Scope of supply

Stationary dry installation - vertical (installation type D)

- Pump set complete with power cables
- Base elbow with inspection hole⁵⁴⁾ and fastening elements
- Optional: intake elbow with inspection hole

Stationary dry installation - horizontal (installation type H)

- Pump set complete with power cables

- Foundation rails
- Suction-side flanged spacer with inspection hole⁵⁵⁾ (optional)

Stationary wet installation (installation types K and S)

- Pump set complete with power cables
- Claw with sealing and mounting elements
- Lifting rope, lifting chain or lifting bail (optional)
- Mounting bracket with mounting elements
- Base elbow and mounting elements
- Guide cable / guide rail
(guide rails are not included in KSB's scope of supply)

Transportable model for wet installation (installation type P)

- Pump set complete with power cables
- Foot plate or pump stool with mounting elements
- Lifting rope, lifting chain or lifting bail (optional)

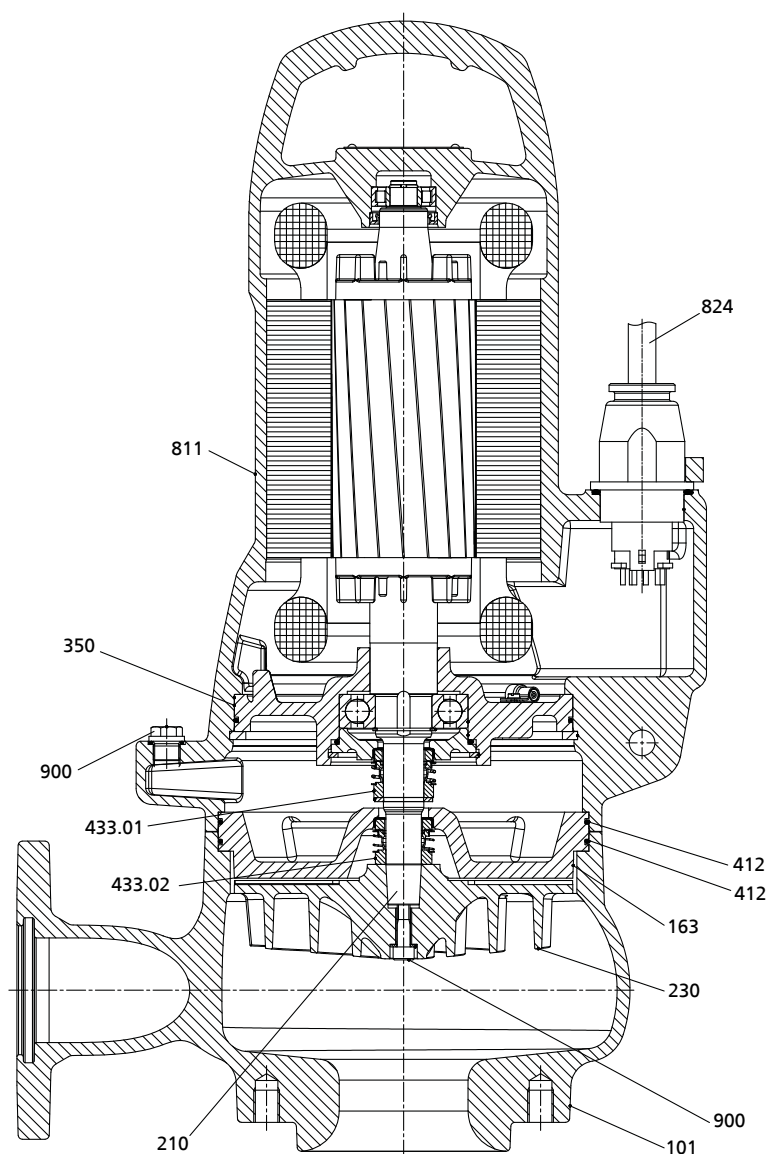
⁵⁴⁾ For nominal discharge nozzle diameter $\geq$ DN100

⁵⁵⁾ For nominal discharge nozzle diameter $\geq$ DN 100

General assembly drawings with list of components

Amarex KRT, motor type 1

Correlation of motor and motor type: (⇒ Page 23)



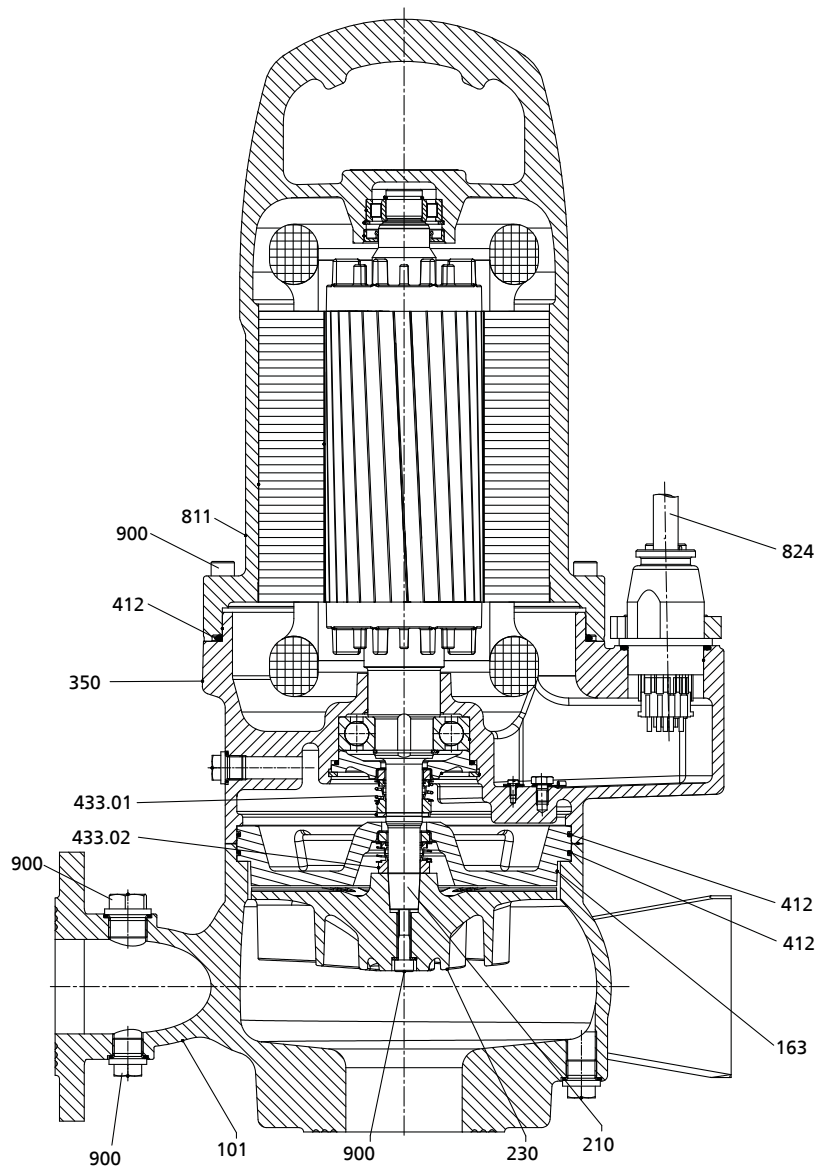
General assembly drawing, example: Amarex KRT F65-215

List of components

Part No.	Description	Part No.	Description
101	Pump casing	412	O-ring
163	Discharge cover	433.01/.02	Mechanical seal
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Screw

Amarex KRT, motor type 2

Correlation of motor and motor type: (⇒ Page 23)



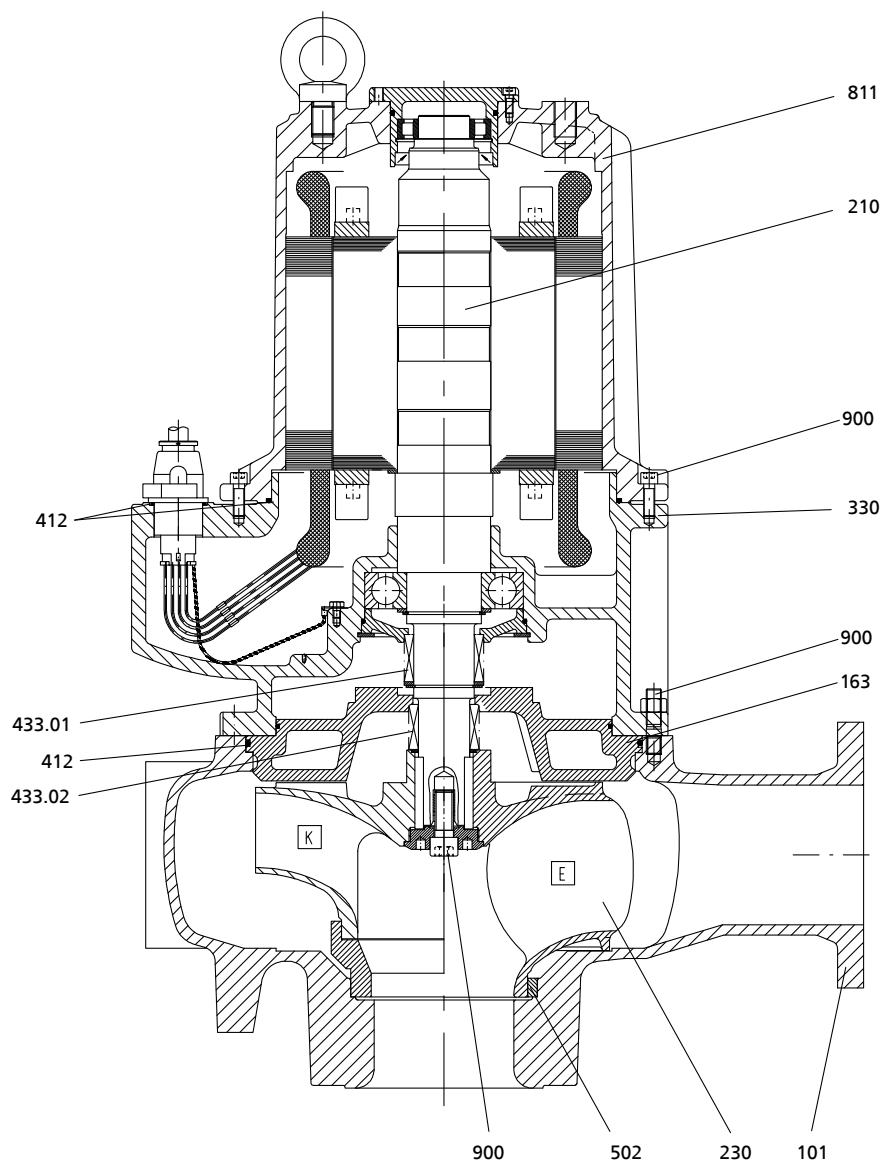
General assembly drawing, example: Amarex KRT F 65-215

List of components

Part No.	Description	Part No.	Description
101	Pump casing	412	O-ring
163	Discharge cover	433.01/.02	Mechanical seal
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Screw

Amarex KRT, motor type 3

Correlation of motor and motor type: (⇒ Page 23)



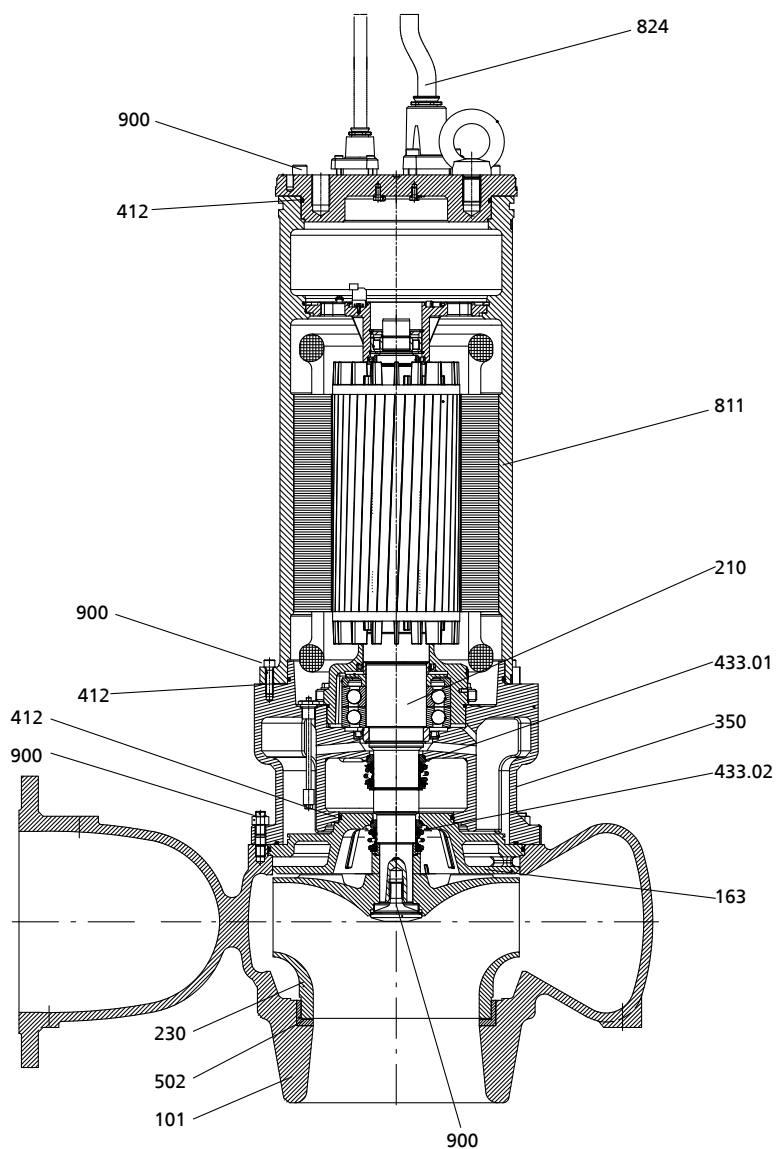
General assembly drawing, example: Amarex KRT E/K 150-401/65 4 XG

List of components

Part No.	Description	Part No.	Description
101	Pump casing	412	O-ring
163	Discharge cover	433.01/.02	Mechanical seal
210	Shaft	502	Casing wear ring
230	Impeller	811	Motor housing
330	Bearing bracket	900	Screw

Amarex KRT, motor type 4, installation types S and P

Correlation of motor and motor type: (⇒ Page 23)



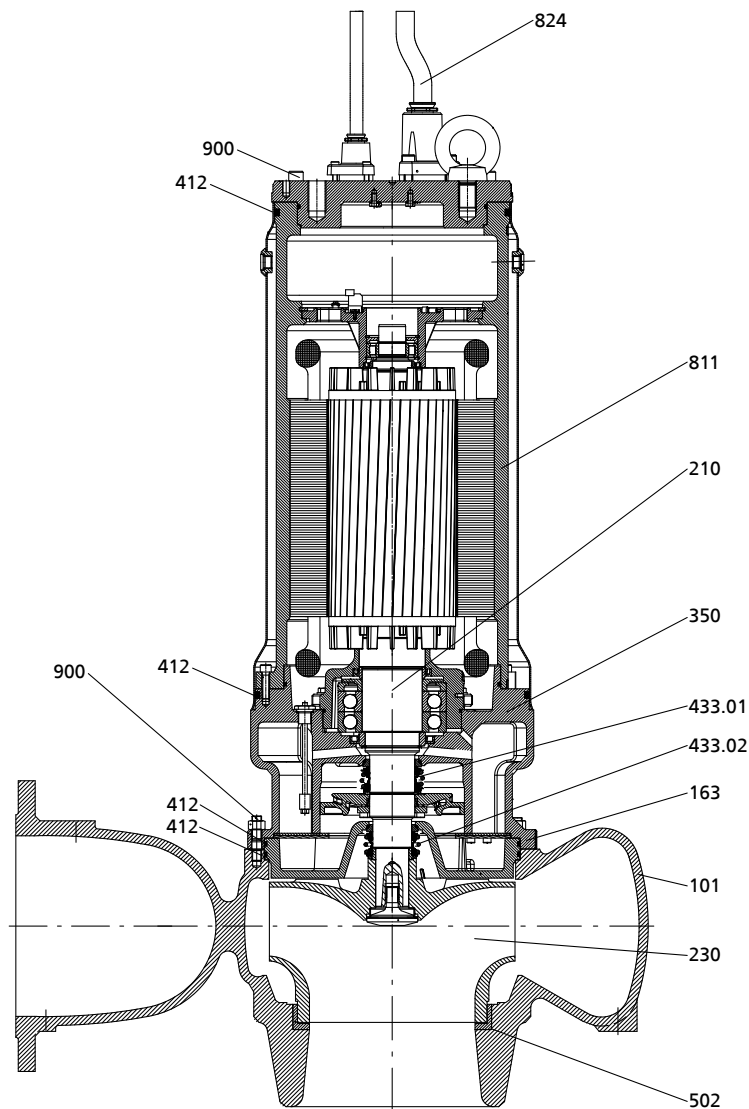
General assembly drawing, example: Amarex KRT K 150-401 / 130 4 XNG-S without cooling jacket

List of components

Part No.	Description	Part No.	Description
101	Pump casing	433.01/02	Mechanical seal
163	Discharge cover	502	Casing wear ring
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Screw
412	O-ring		

Amarex KRT, motor type 4, installation types K and D

Correlation of motor and motor type: (⇒ Page 23)



General assembly drawing, example: Amarex KRT K 150-401 / 130 4 XNG-K with cooling jacket

List of components

Part No.	Description	Part No.	Description
101	Pump casing	433.01/.02	Mechanical seal
163	Discharge cover	502	Casing wear ring
210	Shaft	811	Motor housing
230	Impeller	824	Power cable
350	Bearing housing	900	Screw
412	O-ring		



KSB, Inc.

4415 Sarellen Road • Richmond, VA 23231

Phone: (804) 222-1818 • Fax: (804) 226-6961 • www.ksbusa.com

KSB PUMPS Inc.

5885 Kennedy Road • Mississauga • Ontario L 4Z 2G3

Phone: (905) 568-9200 • Fax: (905) 568-3740 • www.ksb.ca